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Chapter 24

The Evolution of Local Southeast Asian Astronomy and the Influence of China, India, the Islamic World and the West



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24.1 Introduction (YÔ and WO)

There are several traditional calendars in Asia, and in some regions, they are still used for traditional festivals etc.

For example, there are several Thai traditional calendars that are popular today (see Fig. 24.1), and in 2017 the first author of this chapter also was able to purchase some Burmese calendars at a bookshop in Mandalay where Burmese traditional dates are mentioned (Fig. 24.2).

There are several traditional festivals in Thailand and Myanmar, such as *Songkrān* in Thailand, which corresponds to *Thingyan* in Myanmar, and *Lōy Krathong* in Thailand, which corresponds to the *Tazaungdaing* festival in Myanmar, etc. (for Thai festivals see Gerson, 1996; Kajiwara, 2013).

Songkrān was originally a Sanskrit word *saṅkrānti*, which means the Sun's entrance to a sign of the zodiac. The famous Thai *Songkrān* festival is the Sun's entrance to the sign Aries, that is the vernal equinox. However, *Songkrān* currently

Unfortunately, Dr Yukio Ôhashi died suddenly on 31 October 2019, so this chapter combines his original manuscript and additional material provided by Wayne Orchiston. The editors of this book felt it essential that Yukio's chapter be included in the book, in order to round out the discussion and close off the book. But there were legal issues, and the only way we could address these was by adding a co-author and substantially expanding Yukio's original manuscript. To indicate authorship, the following code is used after each heading: (YÔ) = prepared by Yukio Ôhashi; (WO) = prepared by Wayne Orchiston; and (YÔ and WO) = prepared by both authors.

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Fig. 24.1 Recent Thai traditional calendars (Ôhashi Collection).

is celebrated during 13–15 April in Thailand, because the Indian sidereal year is used for the traditional Thai calendar. Meanwhile, according to Ma Thanegi (2014: 51), Burmese New Year falls on 17 April, and *Thingyan* is celebrated for five days prior to that date. *Lōy Krathong* is on the Full-Moon day of the 12th month of the traditional Thai luni-solar calendar.

We shall come back to the relationship between the Indian sidereal year and lunar months in the traditional Thai calendar in Sub-section 24.2.2, later in this chapter.

In several areas of Asia, traditional astronomy has a ‘multiplex structure’ consisting of local original astronomical knowledge plus the influences of China, India, the Islamic World and the West. For example, Tibetan traditional astronomy (Ôhashi, 2000) and related Mongolian astronomy are a kind of multiplex astronomy consisting of local knowledge, Indian Classical astronomy, certain information derived from the Islamic calendar and Chinese traditional astronomy. Iran also was a kind of crossroads of traditional astronomy, as indicated by some recent studies (e.g. see Isahaya, 2009, 2013).

In this chapter, we shall see that there are several different kinds of multiplex astronomy in Southeast Asia. This long chapter is a substantially revised and greatly expanded version of the papers that the first author (YÔ) presented at the meeting of the Southeast Asian Astronomy Network’s History & Heritage Working Group on “Researching the History of Astronomy in Southeast Asia” in Ao Nang, Thailand in 2015 and the Working Group’s second meeting in Mandalay, Myanmar in 2017, along with a range of additional material provided by the second author (WO). After



Fig. 24.2 Recent Burmese calendars (Ôhashi Collection).

the Ao Nang meeting, YÔ also presented an overview of Southeast Asian astronomy in Japanese in Tokyo, and later this was published (Ôhashi, 2016b). After the Mandalay meeting, he also presented a paper in Japanese (Ôhashi, 2018) at a research seminar in Japan. This chapter also draws freely on some of the material prepared by WO when researching Chapters 1, 2 and 19 in this book.

24.2 Geographical Overview: Mainland Southeast Asia

Mainland Southeast Asia can be divided into Vietnam, where Chinese astronomical influence was strong, and other areas, where Indian astronomical influence was strong. For the history of Mainland Southeast Asian astronomy in general see Ôhashi (2005, 2008, 2011).

24.2.1 Vietnam (YÔ and WO)

Chinese classical calendars were used in Vietnam. Sometimes, Vietnam continued to use the old Chinese calendar even after the introduction of a new calendar in China, which led to certain differences between Vietnamese and Chinese astronomy. Most pre-modern Vietnamese astronomical works were written in Classical Chinese. Fig. 24.3 is an example of a manuscript on Vietnamese astronomy (*Thiên-văn khảo* (天文考, *Study of Astronomy*) written in classical Chinese language, which is Volume 1 of the *Sử-học bị-khảo* (史学備考) of Đặng Văn-phủ (鄧文甫) and is preserved in the Institute of Sino-Nom Studies in Hanoi.

From around CE 1300 to CE 1812, the Chinese theory of the Shoushi-li (授時曆) Calendar and its successor, the Datong-li (大統曆) Calendar were basically used in Vietnam, but it seems that the calculations were done in Vietnam. Even after the Shixian-li (時憲曆) Calendar was adopted in China, the Datong-li Calendar continued to be used in Vietnam until CE 1812 (Ôhashi, 2016a).

From CE 1813, the Shixian-li Calendar was adopted in Vietnam. At the time of Emperor Minh-mạng (明命) (reign 1820–1840) of the Nguyễn Dynasty, calendar reform was carried out, and it was declared in 1837 that the prime meridian for Vietnam passed through its capital, Hue. After 1841, the longitudinal difference between Beijing and Hue was taken into consideration during the calculation of the calendar. So, from that time on, the Vietnamese classical calendar differed from the Chinese classical calendar (Ôhashi, 2004, 2005).

After World War II, before the reunification of Vietnam, North Vietnam decided to use UTC+7 as standard time in 1968, while South Vietnam used UTC+8, which was the same as Chinese standard time. Since the reunification in 1975, UTC+7 has been used. So, Vietnamese traditional New Year's Day *Tết* sometimes differs from Chinese traditional New Year's Day. For example, Chinese traditional New Year's Day in 1985 was 20 February, but Vietnamese traditional New Year's Day was 21

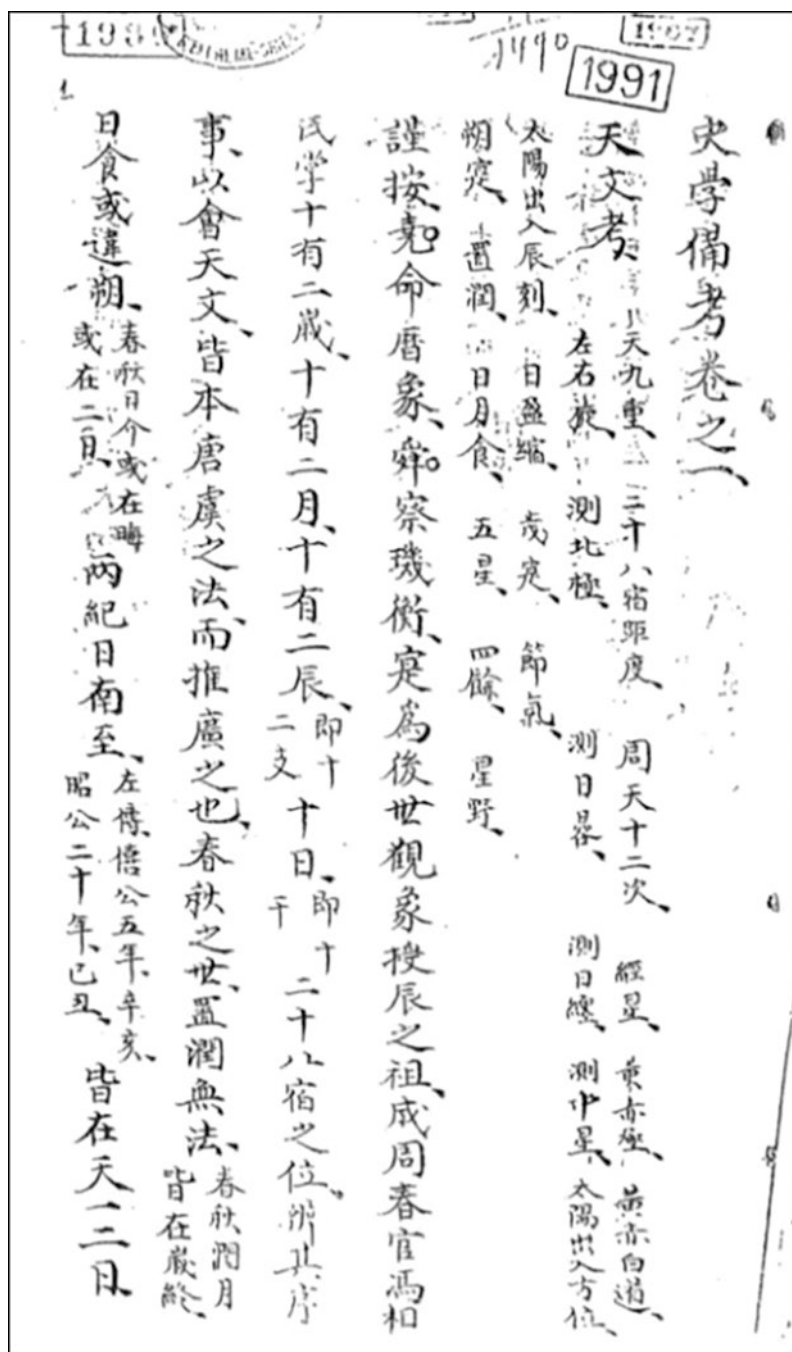


Fig. 24.3 A manuscript of a work on Vietnamese astronomy (courtesy: Institute of Sino-Nom Studies, Hanoi).

January (Takeuchi, 1987). There are papers about traditional Vietnamese calendars and Vietnamese astronomy by Lê (2010, 2019), Lê and Truong (2017), Okazaki (2017, 2021) and Phâm and Lê (2021), while Nguyen and Nguyen (2004) report on the bamboo stick calendar of the Muong people of Vietnam.

We also need to consider the Champa Kingdom, which existed in present-day central and south Vietnam until 1932. It was much influenced by Indian culture, and dates in inscriptions using the Śaka Era were mentioned in Indian style from the sixth century onwards (see Sugimoto, 1956). The present-day *Cham* people in Vietnam are Austronesian speakers, are said to be the descendants of the Champa Kingdom, and have their own distinctive calendrical system (see Nakamura, 1999, 2009; Sakaya, 2016; Yoshimoto, 2000, 2003, 2011). Their calendars are discussed further in *Sub-subsection 24.9.3.1* below.

24.2.2 *Thailand, Laos, Cambodia and Myanmar (YÔ and WO)*

Traditional astronomy in this area is a combination of local indigenous astronomy, Chinese influence and Indian influence. For traditional calendars in this area see Eade (1995) and Gislén and Eade (2019a, 2021), and for Indian astronomy viewed from a Southeast Asian perspective see Ôhashi (2009, 2016b).

The *Tai* (傣, *Dai* in Chinese pinyin transliteration) people in Sipsong-panna (西双版纳, Xishuang-banna in pinyin transliteration) in Yunnan (云南) Province of China are closely related to the Thai people, and have traditional astronomical texts. There are detailed studies of the astronomy of the *Dai* people by Zhang and Chen (1981) and Zhang (2013: Volume 3). The traditional culture of the *Ahom* people (one of the *Tai* peoples) in Assam, India, is also close to this area. For their astronomical knowledge, see Terwiel and Wichasin (1992). The *Cham* people (mentioned above) also are culturally similar.

In Thailand, Chinese influence and Indian influence can be seen in an early inscription. In the inscription of King Ram Khamhaeng of the Sukhothai Dynasty of Thailand (see Chamberlain, 1991), a year is mentioned as “In 1214 Śaka, year of the Dragon”, which corresponds to CE 1292. Both the Indian year (1214 Śaka) and the Chinese year (year of the Dragon) are used. Then in the fourteenth century, King Lithai of the Sukhothai Dynasty wrote the *Trai-phūm* (*Three Worlds*) which described Buddhist cosmology in detail (for the English translation of this work see Reynolds and Reynolds, 1982). Then during the Ayutthaya Dynasty de La Loubère (1693) provided a detailed record of Siamese culture, including traditional astronomy, which was examined and explained by Paris Observatory’s first astronomer Jean-Dominique Cassini (1625–1712). This marked the first time that information about an Indianized calendar was introduced to European astronomers. For information about the traditional Thai calendar see Gislén and Eade (2019b, 2019d), Komonjinda et al. (2017), Nonaka (1987) and Suehiro (1987).

Some different eras were used in Thailand. In the Sukhothai inscriptions, the Śaka Era, which is called *Mahā-sakkarāt* in Thai, was used, of which the best known

is that of King Ram Khamhaeng. Its 0th year began in CE 78 (so, CE 78/79 = *Mahā-sakkarāt*). Then, *Junla-sakkarāt*, which is the same as the Burmese Era, was used through to March of CE 1889. Its 0th year began in CE 638 (so, CE 638/639 = *Junla-sakkarāt*). Until March of CE 1889, the luni-solar calendar was used officially in Siam. Since 1889, the Gregorian Calendar has been used, but each year began from April (instead of January) up until 1940. From April 1889 to March 1912, the Rattanakosin Era, or *Rattanakōsin-sok* in Thai, was used. Its 1st year began on CE 1782 (so, CE 1781/1782 = *Rattanakōsin-sok*). From April 1912, the Buddhist Era, which is called *Phuttha-sakkarāt* in Thai, has been used. Its 0th year began in 544 BC (so, CE + 542/543 = *Phuttha-sakkarāt*). From 1941, the Thai year began in January, and *Phuttha-sakkarāt* has been commonly used until now (now, CE + 543 = *Phuttha-sakkarāt*). Eade (1996) gives several examples of dated inscriptions and chronicles using *Mahā-sakkarāt*, *Junla-sakkarāt* and *Phuttha-sakkarāt*.

For Thai folk-astronomy, see Maison (2013), Phengkaew (2009) and Saibejra (2012). There are some Thai original asterisms (see Sub-subsection 24.4.1.1 below). The National Astronomical Research Institute of Thailand (NARIT) has published a series of books on traditional Thai astronomy in the Thai language, namely, Phumphonphaet (2013) on Thai archaeoastronomy, Maison (2013) on Thai folk-astronomy, and Sawasdee and Maison (2013) on Thai calendrical astronomy.

There was an independent Lanna Kingdom in northern Thailand from the end of the thirteenth century to the beginning of the twentieth century, and the *Lanna* people had their own astronomical systems and beliefs which were based on Indian astronomy (see Soonthornthum 1998, 2006). For the northern Thai calendar, see Davis (1976).

There are a number of Chinese ethnic minorities found in the mountainous regions of northern and north-western Thailand that have their own distinctive astronomical systems. According to Iijima (1971: 81 and 109), who studied the *Karen* people of northern Thailand in 1963–1964 and 1964–1965, the *Karen* calendar begins from the month *Tale* (mid-December to mid-January in the Gregorian Calendar), and the months *Teku*, *Tepe*, *Lasa*, *Denya*, *Lanwe*, *Laxo*, *Laku*, *Chimú*, *Chicha*, *Lano* and *Laplu* follow.

The *Mon* people are an ethnic minority group with Myanmar origins who are now also resident in central Thailand, and speak the Mon language which is one of the Austro-Asiatic languages. The *Mon* have their own literature, including astronomical texts. When King Rama IV (r. 1851–1868; Soonthornthum and Orchiston, 2021) predicted the total solar eclipse in 1968, he used astronomical texts written in Siamese, Mon and English. According to Thongchai (1994), King Mongkut used a Mon text *Saram*, one of the two Mon treatises for planetary calculation known in Siam; the other text, more conventionally used by astrologers at that time, was *Suriyayat*. Komonjinda et al. (2017) state that King Rama IV created the *Pkakanana* system of luni-solar calendar following the *Saram*, which is complicated but more accurate than the usual Thai luni-solar calendar that follows the *Suriya-yatra*. According to Komonjinda et al. (*ibid.*), the *Pkakanana* system is now used only in *Dhammayuttika Nikaya*, which is an order of Theravada Buddhism in Thailand.

Lao people are closely related to the Thai people, and their traditional luni-solar calendar is described by Dupertuis (1981) and Phetsarath (1940, 1959).

In Cambodia there are inscriptions using the Indian Śaka Era dating from the seventh century CE (Barth, 1885). There is a well-known inscription dated Śaka 605 (= CE 683), which is one of the earliest inscriptions that have numerals with the symbol for zero (Aczel, 2015; Cœdès, 1931; Hayashi, 2018: 31).

There is a Chinese record of the Angkor period of Cambodia titled *Zhenla-fengtuji* (真臘風土記) written by Zhou Daguan (周達觀), who in CE 1296–1297 visited Cambodia (which was called *Zhenla* in Chinese at that time). Zhou's account (2000) includes a section on the calendar: the first month of the Cambodian calendar corresponded to the tenth month of the Chinese calendar and was called *jiade* (佳德). However, this term *jiade* originated from the Sanskrit word *Kārttika*, so we can see the influence of the Indian calendar in Cambodia.

According to Porée and Maspero (1938), there were four astrologers called *hora* in the Cambodian court, and they made calendars and predicted auspicious times etc. For details of Cambodian astronomy see the monograph by Faraut (1910).

The Kingdom of Pagan, the earliest kingdom of the Burmans, existed in present-day Myanmar from the mid-eleventh century to the thirteenth century CE. Luce (1969-1970) tells how Pagan became the Buddhist capital of a united Burma, including Mons and Burmans, from about CE 1060, and he discusses the Old Burma Calendar used in Old Mon and Old Burmese inscriptions dating to the Buddhist, Śaka or Burmese Eras (Luce, 1970(II): 327–337). There are early records of Burmese culture, including cosmology, by Sangermano (1833/1893) and Shway Yoe (1882/1910), while Leider (2005-2006) discusses the function of court astrologers during the Konbaung Dynasty (i.e. from 1752 to 1885).

Burma has its own asterisms, which are described by Buchanan (1799: 195–202), and partially in Khin Zaw (1937). There are some star maps in Burma, the most interesting being three drawn on the ceilings of corridors in the Kyauktawgyi Pagoda at Amarapura, near Mandalay, which was built in 1847 (see Nishiyama 1997). A detailed discussion of these star maps is presented below in Sub-subsection 24.4.1.2.

For information about the traditional Burmese calendar and astronomy see Clancey (1906), de Silva (1914), Gislén and Eade (2019b, 2019d, 2021), Htoon-Chan (1918), Irwin (1909) and Kiryū (1987), while Furnivall (1922) discusses the Indian Jovian 12-year cycle in Burmese inscriptions. In the traditional Burmese calendar the Burmese Era is used, where the 0th year corresponds to CE 638. Note that the Burmese Era corresponds to the *Junla-sakkarāt* of the Thai calendar.

Irwin (1909) reports the difference between the Burmese calendar and the Arakanese (Rakhine) calendar in the early twentieth century. The Burmese calendar used the *Makaranta*, which probably followed the original *Sūrya-siddhānta*, until the nineteenth century. The *Thandeikta*, which chiefly followed the present *Sūrya-siddhānta* and is said to have been composed in about CE 1738 or CE 1838, was used in the Burmese calendar thereafter. Meanwhile, the *Makaranta* was still used by the Arakanese. There have been recent studies of Burmese astronomy by Gislén (2015, 2018) and Gislén and Eade (2014).

24.2.3 *The Malay Peninsula (YÔ and WO)*

Early records of astronomical knowledge in the Malay Peninsula were collected by Skeat (1900: 532–566). According to this work, the Malay people in the Malay Peninsula used a 5-day cycle and a 7-day week. The signs of the zodiac had Arabic names. One lunar month was divided into days that seemed to have originally corresponded to the Indian lunar mansions, but their number was usually raised to 30 so as to fit a lunar month. Both the solar year (= tropical year) (365 days) and the lunar year (= 12 synodic months) (354 days) were used. For the calculation of the Islamic lunar year, the 120-year cycle (= 1440 synodic months = 42,524 days) and the 8-year cycle (= 96 synodic months = 2835 days) were used. They were the usual simplified methods for calculating Islamic lunar years.

In modern Malaysia, there is a controversy regarding the Islamic calendar. According to Horii (1987), who studied in Malaysia, some people prefer the actual observation of the crescent Moon (*rukayah*) while others prefer calculations (*kiraan falak*) to determine the first day of an Islamic month. For example, Horii reports that in August 1981, the Sultans of Perak State and Johor State decided the day of the festival Hari Raya Aidilfitri (the first day of the month Syawal after the end of the month Ramadan) by actual observations of the crescent Moon, and it was one day earlier than in other states in Malaysia.

The indigenous minorities in the Malay Peninsula are called *orang asli* (the original people) and consist of the ‘Semang’ (negritos), ‘Senoi’ (who speak Austroasiatic languages) and Proto-Malays. Their cultures are described by Skeat and Blagden (1906). According to this work, the Proto-Malay people denote seasons by the north and south monsoons, and also the harvesting of rice and the ripening of fruits. They know time by the inclination of a stick directed to the Sun (Skeat and Blagden, 1906(I): 393). Semang people think that the Sun and the Moon are female, and each has a husband, while the stars are the Moon’s children. Eclipses are attempts by a gigantic serpent or dragon to cover or swallow the Sun, and there is great fear (Skeat and Blagden, 1906(II): 202–203). Recent studies of indigenous Malaysian astronomical systems are reported elsewhere in this book (see Jaafar and Khairuddin, 2021a, 2021b; Khairuddin and Jaafar, 2021).

24.3 Geographical Overview: Island Southeast Asia (YÔ and WO)

Traditional astronomy in Island Southeast Asia is based on local original astronomical knowledge and Indian, Islamic and sometimes Western influences.

For information about the Indianized calendar in this region see Casparis (1978), Damais (1967), and Eade and Gislén (2000). In Java, there are inscriptions dated by the Indian Śaka Era from the eighth century (Nakada, 1982), and there is also an inscription dated to Śaka 605 (= CE 683) in Sumatra, which is one of the earliest

inscriptions that has numerals with the symbol for zero (Cœdès, 1931; Hayashi, 2018: 31). For details of the Islamic calendar in this area see Gislén and Eade (2019c) and Proudfoot (2006). For the astronomy of this region in general see Ammarell (2008), Hidayat (2000) and Maass (1924–1926).

Although the Andaman and Nicobar Islands belong politically to India, their traditional cultures are close to those of island Southeast Asia. For information on the astronomical knowledge of the negritos of the Andaman Islands see Radcliffe-Brown (1928: 141–150), and the indigenous groups of the Nicobar Islands see Man (1897; Vahia et al., 2018).

There also were Austronesian people in Taiwan who were culturally close to the inhabitants of Island Southeast Asia, and there were several studies made by Japanese ethnologists early in the twentieth century (see Ôhashi, 2017b).

We also should note that there are cultural links between Island Southeast Asia and the Polynesian, Melanesian and Micronesian inhabitants of the Pacific islands (e.g., see Bellwood, 2017; Carson et al., 2013).

24.3.1 *Sumatra (YÔ)*

Marsden (1811: 193–194) records that the Malays used the Islamic calendar (where one year is 354 days), but the original Sumatrans estimated their annual periods from the revolution of the seasons. To denote the time of day, they pointed with their finger to the height of the Sun. They noticed the planet Venus, but did not imagine that the morning Venus and the evening Venus were the same. During an eclipse, they made a lot of loud noise with percussion instruments to frighten off the snake or dragon intent on devouring the Sun.

In Sumatra, there are Batak and Nias peoples etc. (who are Proto-Malays), and Acehnese, Minangkabau and Lampung peoples, etc. (who are Malays).

There is a detailed work on Acehnese culture by Hurgronje (1906, cf. Kurata, 1982), and the Acehnese calendar is described on pages 194 ff. According to this work, the Islamic lunar year is used by Acehnese people (who live in northern Sumatra). Although the commencement of each month should basically be determined by observation, in many districts the calculation is adhered to, and the 8-year cycle and the 120-year cycle are used. There are three intercalary years (355 days) in eight years. In this method, there is a 1-day excess every 120 years, but Hurgronje was not been able to discover whether the Acehnese people corrected for this at the end of every 120 years by skipping a day. The Acehnese are agricultural people, and a seasonal calendar is needed. For this purpose, stars are used. Acehnese people know the three stars in Orion's Belt, Venus (they consider the 'Morning Star' and the 'Evening Star' to be two different stars), the Southern Cross, the Scorpion, the Pleiades, etc. The line joining the three stars in Orion's Belt was thought to point to Qibla, and this idea also prevails in Java. Conjunctions of Scorpion and the Moon are considered to regulate the seasons—there are 13 (sometimes 14) conjunctions in

a solar year. Sometimes conjunctions of the Pleiades and the Moon are also used. The Acehnese people used the N.E. and S.W. monsoons for navigation.

Kiyono (1943: 492–497), referring to the works of the German ethnologist A. Maass, mentions some astronomical knowledge of the Minangkabau people of central Sumatra. Time was determined from the altitude of the Sun. At 1 p.m. a vertical rod was erected, and the prayer ‘Asr’ had to be completed before the length of the shadow exceeded the height of the rod. One month is from a New Moon to the next New Moon, as per the Islamic calendar. The seasons were known from the south, east, west and north monsoons, and there are also rainy (wet) and dry seasons. The Minangkabau people knew the Morning Star, Evening Star, the three stars in Orion’s Belt, and the Scorpion, while specialists also knew the Pole Star, comets, planets, the Great Bear, Virgo, the Magellanic Clouds, and more.

The Lampung people in southern Sumatra used the usual Islamic calendar, and they also had a 30-day month (Ginzel, 1906: 426ff).

According to Loeb (1935: 139–140), the people who lived on the island of Nias (off the northwest Sumatran coast) begun their year with the rising of the Pleiades, and planting began one month later. A year was divided into 12 numbered lunar months, and a month was divided into the waxing and the waning of the Moon. There was a story that seven children (the Pleiades) went up to the sky, and their parents and a slave (Orion) followed. There was also a story that at one time there were two Suns and no Moon, and the stars were the children of the Sun and Moon. According to Loeb (1935: 172–173), people living on the Mentawai Islands also used the position of the Pleiades to mark the beginning of the new year in June.

The Batak people who live around Lake Toba have an interesting traditional calendar (see Kimball, 1989–1993; Shinotsuka, 1987; Winkler, 1913/1989–1993). The Batak calendar is called *porhalaan* and is engraved on a bamboo cylinder. There are 12 or 13 columns (for the months) by 30 row rectangular grids (for the days). A year begins when Orion disappears in the western sky and Scorpio rises in the eastern sky. When the rising crescent Moon passes Orion immediately after sunset the Batak year begins. Although modern Batak people use Indonesian day-names of the week that are of Arabic origin, the traditional Batak calendar uses day-names of the week that are of Sanskrit origin.

24.3.2 *Java and Madura (YÔ and WO)*

In Java, there are Javanese people in central and eastern Java, Sundanese people in western Java, and Madurese people in eastern Java and on the island of Madura. They are all Malay people.

There are several dated inscriptions in Java that use the Indian Śaka Era (see Damais, 1967; Eade and Gislén, 2000; Nakada 1982) while Raffles (1817: 473–479) and Ginzel (1906: 414ff) give early descriptions of astronomy in Java. Igarashi (1987) gives a detailed description of the Javanese calendar (cf. Takahashi, 1987), and Geerts (1960: 77–81) discusses its religious aspects.

The Javanese lunar calendar was established by Sultan Agung (the third King of the Mataram Sultanate) who reigned between 1613 and 1645. It is a combination of the traditional Javanese calendar and the Islamic calendar. The day begins at sunset. Besides the 7-day week, the 5-day cycle *pasaran* is used. One year of the Javanese calendar is a lunar year, and the 8-year and 120-year cycles are used for calculation. One usual year is 354 days, but there are three 355-day years every 8 years. And also, one day is omitted in every 120-year cycle. For counting years, the Indian Śaka year is used instead of the Islamic Hijri year, and CE 1633 was Javanese year 1555. As one lunar year is shorter than one solar year, the Javanese year is now different from the Śaka year of the solar calendar.

In CE 1855 the traditional Javanese solar calendar was reformed by Pakubuwono VII, a ruler of Surakarta who reigned from 1830 to 1858. It begins from the summer solstice, and there was considerable variation in the length of each month. This system of calendar is called *Pranotomongso* (or *Pranatamangsa*), and was linked to certain asterisms (Dardjoeni and Hidayat, 1987). There is a detailed discussion about *Pranotomongso* by van den Bosch (1980). One year consists of 12 months, which consist of 41, 23, 24, 25, 27, 43, 43, 26 or 27, 25, 24, 23 and 41 days respectively. This calendar is based on observations made using a gnomon like the one shown in Fig. 24.4a (Maass, 1924-1926; see also Ammarell, 2008: 326; cf. Fig. 24.4b

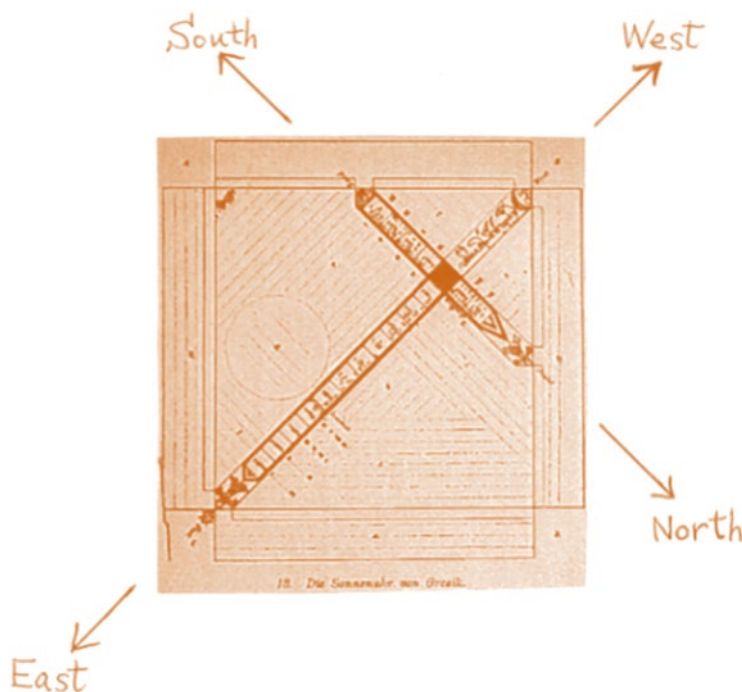


Fig. 24.4a Gnomon from Gresik Regency in Java (after Maass, 1924-1926: Figure 13, with modifications by Yukio Ôhashi).

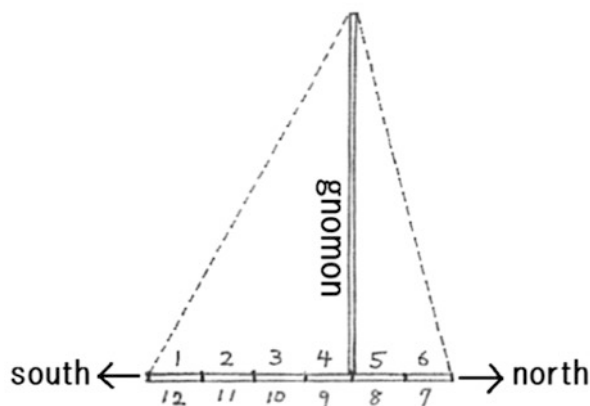


Fig. 24.4b The gnomon used in Java for determining the months in the *Pranotomongso* calendar (diagram: Yukio Ōhashi).

later in this chapter). The noon shadow is observed by its North-South scale. A rectangular gnomon is erected at the square of the intersection of the North-South scale and the East-West rod. The North-South scale is divided into six equal segments (two in the North, and four in the South), and one segment corresponds to a month. As the change of solar declination is not uniform, and the equal segments of the flat scale do not correspond to equal angles, the length of a month is not uniform.

Recently, Ali Sastramidjaja (1935–2009) reconstructed the Sunda Calendar (also called the *Kala Sunda*; see Sastramidjaja 1991/1998). There are separate Sunda solar and lunar calendar systems. In the Sunda solar calendar system, one year ends with the summer solstice, and one month consists of 30 or 31 days. A typical year consists of 365 days, while a leap year has 366 days. A leap year occurs once in four years, but it becomes a common year once in 128 years. In the Sunda lunar calendar system, a half month begins with a half bright Moon and is called *sukla paksa*, and another half month begins from half dark Moon and is called *kresna paksa*. So, one month consists of 29 or 30 days. There is no intercalary month. The 8-year cycle and the 120-year cycle are used to regulate a short year (354 days) and a long year (355 days). The 5-day cycle, 7-day week etc. are also used. Sastramidjaja (ibid.) says that the rules were unknown until he rediscovered them during research that he carried out between 1983 and 1991.

Meanwhile, the inhabitants of the island of Madura, just off the north-eastern coast of Java, ... have their own ideas about the night sky, and they used the appearance or the presence of different stars or groups of stars (asterisms) in everyday life, especially during farming and fishing, or voyaging to other Indonesian islands. The Madurese also had their standard cultural responses to unexpected astronomical phenomena such as solar and lunar eclipses. Before the advent of ‘modern education’, villagers did not understand the scientific reasons for these dreaded events. Even today, many villagers still behave in the traditional way when eclipses occur. (Fatima et al., 2021).

24.3.3 *Bali (YÔ)*

Bali has an interesting tradition of calendar, which is relatively well known. There was a detailed study in 1930's by Covarrubias (1937), and Eiseman (1990, Volume I) also has a detailed description. Eiseman (1990, Volume II) describes star lore in Bali. There are also studies by Japanese: Nagata (1985) and Igarashi (2008). Igarashi (2008) is a very detailed study; also see Emas et al. (2021) and Ginzler (1906, 424–426).

There are two traditional Bali calendars. One is the 'Saka calendar' which was derived from the Indian calendar, and the other is the 'Pawukon calendar' which is a unique calendar introduced from Java.

According to Covarrubias (1937: Chapter 9), the Saka calendar was used by the 'Bali Agas' (the original Bali people). There are some differences in the calendar according to the calendar makers (Igarashi, 2008). In the 'Saka calendar', one day begins from sunrise, and one month begins from the next day of the New Moon. One month is divided into two halves: from the first day of the New Moon to the day of the Full Moon is the white half month (*sukla paksa*), and from the day after Full Moon to New Moon is the black half month (*kresna paksa*). This method is similar to the Hindu calendar. The 0th year of the Saka era is CE 78, and the 1st day of the 10th month (Kadasa) (according to Igarashi, 2008), or the 1st day of the 9th month (according to Covarrubias (1937), is the beginning of a year called 'nyepi' around the vernal equinox. One month consists of 30 lunar days (*titi*), and one day is jumped over every 63 days. One lunar day is 1/30 of a lunar month, and 64 lunar days = 63 civil days in this method. The *titi* corresponds to the *tithi* in the Hindu calendar. For intercalations, there are two methods. One method is to add an intercalary month every 30 months, and the other method is to add 7 intercalary months every 19 years. Now the method of 7 intercalary months in 19 years is used (Igarashi, 2008).

It is said that the 'Pawukon calendar' probably came into use at the time of Majapahit's domination of South Bali (i.e., in the fourteenth century CE). It is made by the cycles of 1 day, 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, 8 days, 9 days and 10 days. One 'wuku' year in this calendar consists of thirty 7-day weeks, that is 210 days.

24.3.4 *The Lesser Sunda Islands (YÔ)*

Bali island (see above) is an island belonging to the Lesser Sunda Islands. There is some information about other islands.

For the calendar of Savu Islands see Cuisinier (1956).

Uno (1941: 120), referring to Dutch literature, tells that in the Ende area of Flores island, when a star called 'red pig' (probably Antares) is seen around altitude 30 degrees from the western horizon around 7 pm, the cultivation season begins. The

‘red pig’ star belongs to an asterism that is considered to be a balance. During this season the Pleiades are seen in the eastern sky.

Maass (1924-1926: 347–357) gives information about astronomy on the island of Timor. Some information about the calendars of Timor and of Letti (now Leti, included in the Moluccas) is mentioned by Ginzel (1906: 430–431).

24.3.5 *Borneo (YÔ)*

The island of Borneo consists of Kalimantan (belonging to Indonesia), Sabah and Sarawak (which are part of Malaysia), and the nation of Brunei. Around coastal areas there are mainly Malay people, while in the mountainous interior there are Proto-Malay people, often referred to as the ‘Dayaks’.

Transilvano, who was a member of Magellan’s voyage (1519–1522), records that when they visited the island of Borneo in 1521 the local people thought that the Sun and the Moon were major gods. The Sun was male and the god of the day-time, the Moon was female and god of the night-time, and the stars were small gods who were related to the Sun and the Moon, and belonging to the Sun and the Moon (Chōnan, 2011: 313). This must be one of the earliest records of the astronomical folklore of Island Southeast Asia.

Hose and McDougall (1912) described some astronomical knowledge in Borneo. They divided the Dayak people into six main groups: the Sea Dayaks (= Ibans), Kayans, Kenyahs, Klemantans, Maryts and Punans. Hose and McDougall (1912(1): 105–111) tell that the Kenyah people erect a vertical wooden gnomon in the ground, whose verticality is ascertained by a plumb, and the length of the midday shadow is measured by a stick with marks on it in order to know the season for agriculture. Some Kayan people observed the position of mid-day sun through a hole in the roof of a chamber. The Klemantan people directed a bamboo cylinder filled with water towards a certain star, and knew the season of planting by the remaining water. The Sea Dayak people knew the seasons by the direction of the Pleiades. Hose and McDougall (1912(2): 139) also mention that the Klemantan people considered that the Great Square of Pegasus was a storehouse; the Pleiades a well; the constellation that included Aldebaran a pig’s jaw; and Orion the figure of a man.

Ammarell and Tsin (2015) report on sky lore in South Kalimantan, where the local people use the three stars in Orion’s Belt to determine the direction of Qibla.

According to Lumholtz (1921(2): 441–444), who carried out research in central Borneo from 1913 to 1917, the Katingan people (one of the Dayak groups in south-west Borneo) used seven inclined gnomons, and when the Sun was above the central gnomon at noon (when the Sun was at or very close to the zenith), they considered that it is the best season for sowing their crops (see Fig. 24.5).

King (1985: 156) records that the Maloh people (who belonged to the Klemantan group of Proto-Malays) also used the stars to determine the times for commencing the agricultural cycle and planting their crops. The most important constellation was

‘bintang tuju’ (Pleiades). The constellation ‘bintang talu’ (Orion) also was recognized.

For the traditional astronomy of the people of northern Borneo see Evans (1922, 1923, 1953).

Freeman (1970: 171–172) reported that the Iban (Sea Dayaks) of Sarawak used ‘bintang banyak’ (Pleiades), ‘bintang tiga’ (Orion) and ‘bintang tangkong peredah’ (Sirius) to track the seasons. The Iban agricultural year was signaled by the rising of the Pleiades at dawn.

24.3.6 *Sulawesi (YÔ and WO)*

On the island of Sulawesi there are Makassar and Bugis people, both of whom are Malays, in the south-west area, Mandar peoples on and near the west coast in the central area (they also are Malays), and the Toraja (Proto-Malay) people in the central area.

Wallace (1869), a noted British naturalist, recorded an interesting navigational aid seen on a local Makassar vessel. It was a floating bowl type of water clock (see

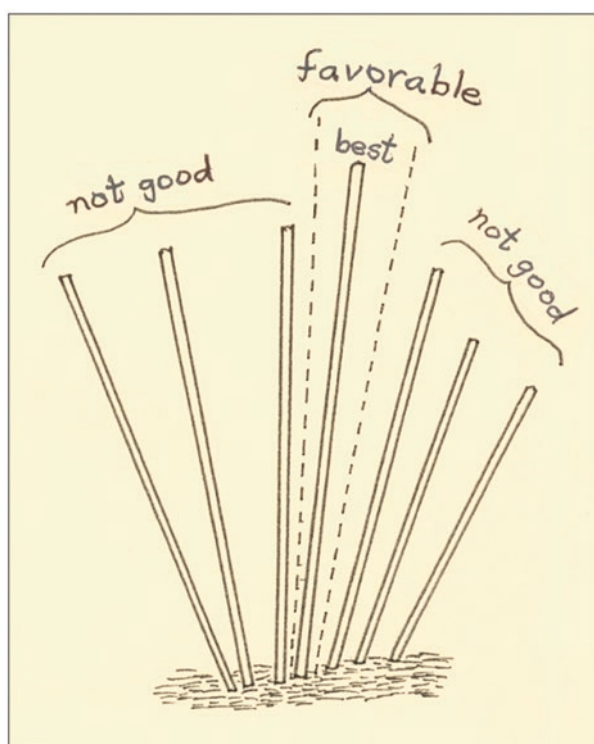


Fig. 24.5 The seven gnomons of the Katingan people in Borneo (after Lumholtz, 1921).

Fig. 24.33). For Bugis navigation, see Ammarell (1999) and Wakita (2010). Ammarell (1999: 127) gives a table of asterisms with identifications. And also, Pelras (1987: 27–32) and (1996: 231 and 264) mentions the constellations of the Bugis people that were used for agriculture and navigation (also see the ‘Cultural Overview’ section below).

Yamashita (1988: 81) records that the Toraja people used stars to track the seasons, and that a year was divided into three seasons. Observation of the stars in order to determine the season was called ‘pentiro taunan’. Stars used included ‘Lemba’ (Ursa Major), ‘Bunga’ (the Pleiades) and ‘Manuk’ (Crux). The three seasons were: the season when these stars were visible in the dawn (i.e., from the end of November until the end of March); the season when they were visible in the evening sky (i.e., from the end of March until the end of September); and the season when they were invisible (i.e. from the end of September until the end of November).

Neighbouring the Bugis on the west coast of Sulawesi is another celebrated maritime group, the Mandar. Just like the Bugis, the Mandar “... developed a body of knowledge involving key stars and star patterns, that traditional navigators passed down from generation to generation.” (Rasyid et al., 2021: 550). In all, ten different stars or asterisms were involved, but they are presently unknown to most young Mandar seafarers. This sad situation is found throughout Southeast Asia, where education and ‘modernisation’ have had a profound impact on indigenous cultures. Consequently, traditional astronomical knowledge is being lost at an alarming rate as old people die without passing on their knowledge to younger generations. As ethnoastronomers we recognize the urgency of our task if we are to capture surviving remnants of these astronomical systems, before it is too late. Hence our motto, inspired by the late great American singer, Elvis Presley, “It’s Now or Never”.

24.3.7 *Moluccas (Maluku Islands) (YÔ)*

In the records of Magellan’s voyage (1519–1522), which was continued after Magellan’s death, there are some records from the Moluccas. Pigafetta writes that the King of Tidore Island was a Muslim and an astrologer (Chōnan, 2011: 166). Transilvano writes that the King said that he knew that the fleet was coming by their observation of the stars (Chōnan, 2011: 324). This was an astrological rather than astronomical observation, but it does show that there was an interest in the stars.

For astronomical knowledge in the Moluccas also see Maass (1924–1926: 364–375).

24.3.8 *The Philippines (YÔ)*

In the present-day Philippines the Gregorian calendar is officially used, but until the end of 1884 the Gregorian day was one day earlier because the Spanish rulers considered that Philippine time was 16 hours later than Spain time. So, 31 December 1884 does not exist in the Philippines, and the next day after 30 December 1884 was 1 January 1885 (Umehara, 1987).

There is some information about traditional calendars in the Philippines in Zaide (1961: 38) and Umehara (1987). According to an early Spanish record dating to CE 1582 the ancient calendar of the Visayans (on the islands between Luzon and Mindanao) had 12 months in a year, among which eight months had names, and the beginning of a year was the time when the Pleiades made their appearance. Zaide tells us that each month had 30 days, except for the last month that only had 26 day, making a year with 356 days. The calendar of the Ifugaos (an ethnic group that lives in the northern part of Luzon Island) contained 13 months (each of 28 days), so one year was 364 days, but a further day was added to the 13th month.

Gōda studied the Bontok people in the northern mountainous area of Luzon Island from the 1970's to 1980's, and he reports on their calendar (Gōda, 1989: 56–71). There, a new year began after the rice harvest, when they were going to start new cultivations, i.e. around October or November. The year was divided into four seasons, and the calendar was not related to the phase of the Moon, although lunar phases were recognized by them.

For the astronomical knowledge found in the Davao district of the island of Mindanao see Cole (1913). Meanwhile, Schlegel (1987–1988) reports on the celestial calendar of the Tiruray, hill people found in Mindanao Island.

For the folk astronomy of the Philippines, see Ambrosio's (2010) monumental work.

24.4 Indigenous Astronomy: A Cultural Overview

24.4.1 *Local Names and Folklore of the Stars and Asterisms (YÔ)*

There are some Southeast Asian groups of stars, or asterisms, which are not found in other areas, and many Southeast Asian asterisms do not correlate with Western constellations.

24.4.1.1 Thai Asterisms

Saibejra (2012), in quoting from the epic *Phra Aphay Manī* written by Sunthorn Phū (1786–1855), mentions the following Thai asterisms:

- Orion: the ‘turtle’ and the ‘plough’ (*Dāw-tau, Dāw-thai*).
- Hyades: the ‘flag’ (*Dāw-thong*).
- Pleiades: the ‘children of the hen’ (*Dāw-lūkkai*).
- Gemini: the ‘coffin’ (*Dāw-lōng*).
- Cassiopeia: the ‘crow’ (*Dāw-kā*).
- Argo: the ‘junk’ (*Dāw-duang-lam-samphau*).
- Big Dipper: the ‘alligator’ (*Dāw-jōrakhē*).
- Arcturus: the ‘summit of great Buddhist tower’ (*Dāw-yōt-mahā-culāmanī*).
- Crux and α and β Centauri: the ‘beam balance’ (*Dāw-khanchang*).
- Great Square of Pegasus: the ‘carrying corpse’ (*Dāw-hām-phī*).

Phengkaew (2009) and Maison (2013) also describe several other Thai astronomical folklores, etc. Fig. 24.6 is an overview of Thai asterisms drawn by the first author of this chapter but based on Saibejra (2012).

24.4.1.2 Burmese Asterisms

Burma has its own asterisms. The earliest description of Burmese asterisms in a Western language is probably by Buchanan (1799, particularly pages 195–205). He says that during his stay in Amarapura, he saw several treatises on astronomy, besides almanacs, including a plan of ‘twenty-seven signs’ and a plan of ‘nine signs’ (see Fig. 24.7), and also a delineation of the sixty-eight Burma asterisms (Fig. 24.8) with a short explanation in the Burmese language.

The small plan in Fig. 24.7 represents the nine northern Burmese asterisms, and the large plan in Fig. 24.7 represents the Burmese northern asterisms (the outer circle) and the 27 Indian lunar mansions (the inner circle).

Fig. 24.8 gives figures of 68 heavenly bodies (the nine northern Burmese asterisms, 27 Indian lunar mansions, other Burmese asterisms, the Sun, Venus and Jupiter). In Fig. 24.8, the first author of this chapter (YÔ) added the blue frame for the nine northern Burmese asterisms, the red frame for 27 Indian lunar mansions, and the yellow background for the Sun, Venus and Jupiter. Following the arrangement of the words in the original, Buchanan gives a verbal English translation, along with a Burmese explanation and his notes. Some asterisms are related to certain cities or countries. Following is a list of Buchanan’s translation of the heavenly bodies (but YÔ has excluded his notes). The first numerals correspond to the numbers given in Fig. 24.8. In Buchanan’s translation, a ‘circle’ means a star. As it is a ‘verbal translation’, the word order is different from the usual English word order. YÔ rearranged and divided Buchanan’s ‘heavenly bodies’ into four groups. The nine northern Burmese asterisms could be identified by their names, and their figures by comparison with Fig. 24.7. It seems that the 27 Indian lunar mansions could not be fully identified with the heavenly bodies in Buchanan’s list, but YÔ could identify them by their names and their figures when compared to the figures of the Burmese lunar mansions quoted by Nishiyama (1997). Most of the Burmese asterisms were not correlated with modern Western constellations by Buchanan.

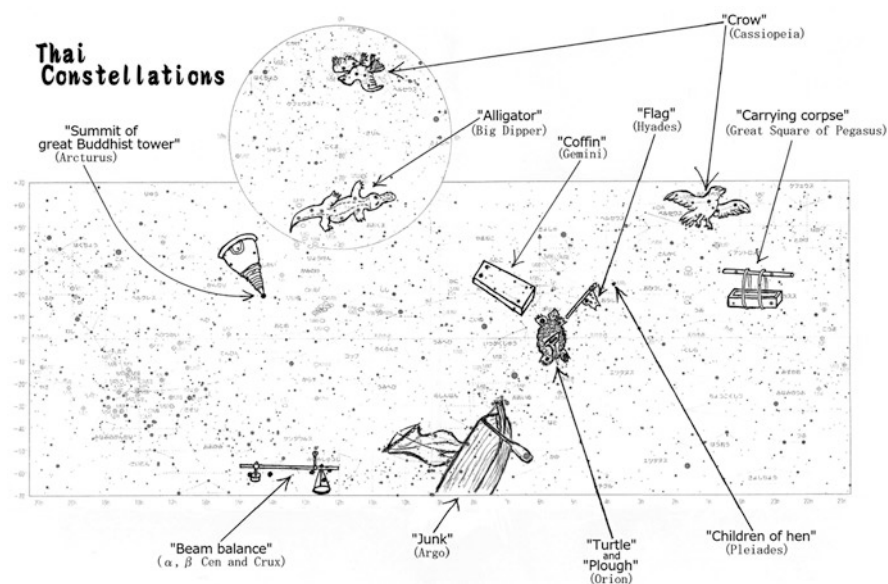


Fig. 24.6 Thai asterisms, drawn by the first author of this chapter, but based on data in Saibejra (2012).

Here is a list of the sixty-eight heavenly bodies mentioned in Buchanan's paper (1799), as re-arranged by YÔ:

(a) The nine northern Burmese asterisms (i.e. 'constellations'). YÔ re-arranged the order in accordance with the prograde order in Fig. 24.7, and he added some notes in brackets. He also added corresponding Roman numerals with his own identifications):

41. "The crow constellation eleven circles has, and the *Thayndua* country" (II).

43. "*Hayntha*, a constellation of seven circles, belong to *Radanapura*". (III)

32. "The crab constellation of ten circles has, *Rasagyol* country". (Buchanan says that no.34 is the Burmese asterism, but no.34 is actually an Indian lunar mansion, and no.32 should be the Burmese asterism.) (IV)

56. "The balance constellation, four circles has". (V)

61. "Of *Thanliæk*, a constellation of three circles, *Kothambe* is the country". (VI)

23. "The *Brahmen* constellation of eight circles, *Kaleingareet* country governs". (Buchanan says that no.15 is the Burmese constellation, but no.15 is actually an Indian lunar mansion, and no.23 should be the Burmese asterism.) (VII)

22. "The *Shan* country the elephant constellation with six circles has". (VIII)

10. "The horse constellation has eleven circles, *Europe* is its country". (IX)

2. "The *Pyain* constellation five circles has, of *Thoukkada* country the constellation". (I)

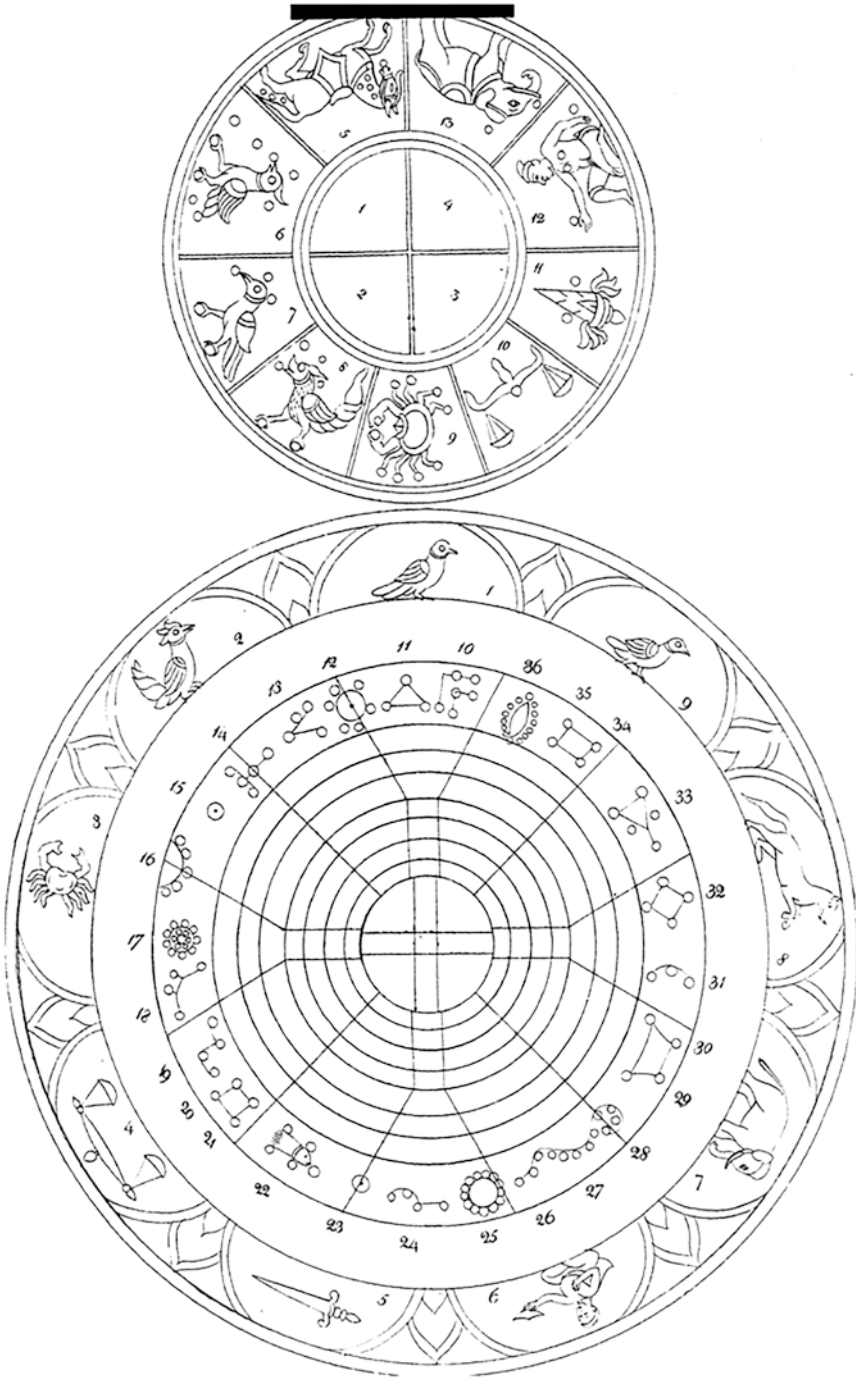


Fig. 24.7 Plans of the 27 Burmese lunar mansions and the nine northern asterisms (after Buchanan, 1799).

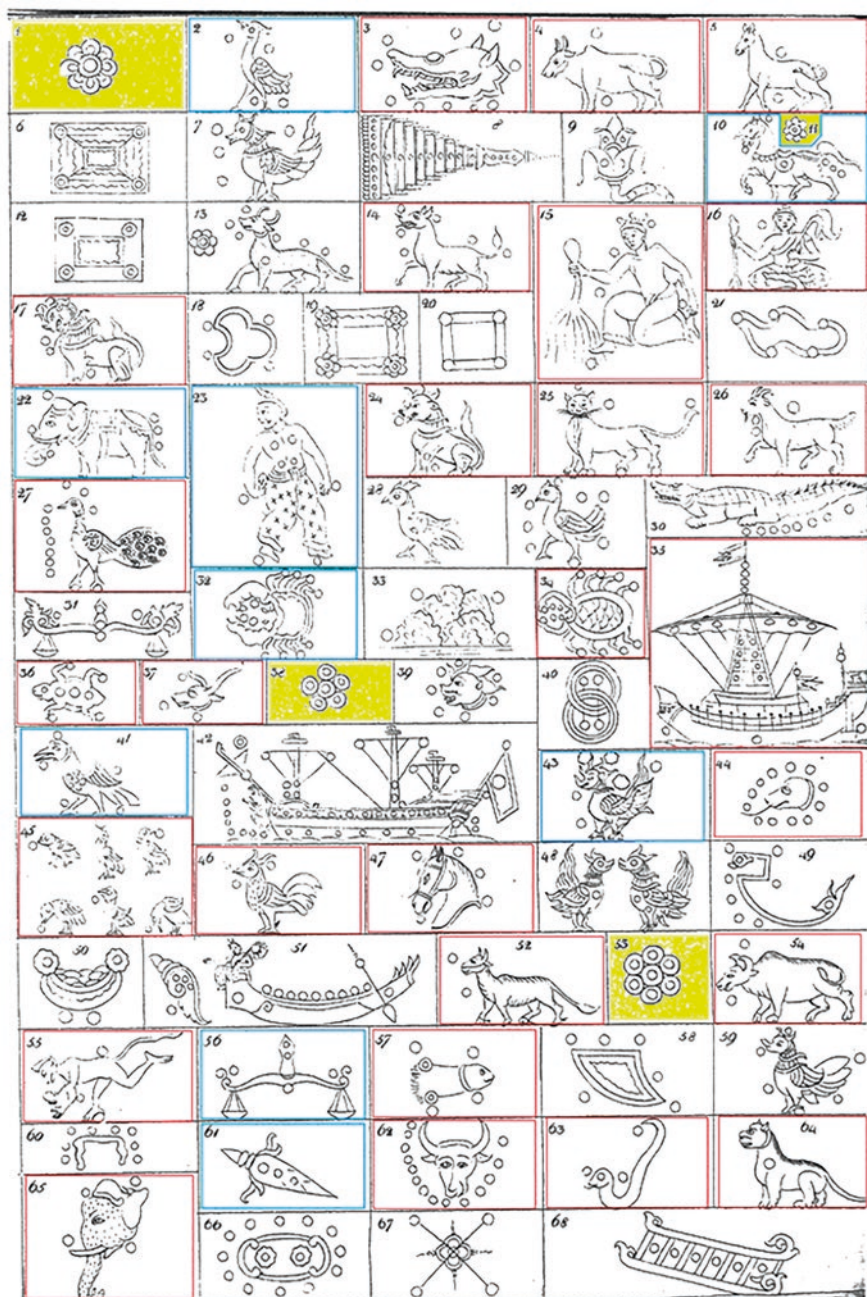


Fig. 24.8 Figures of the 68 Burmese heavenly bodies (after Buchanan, 1799) with the addition by YÔ of blue frames for the nine northern asterisms, red frames for the 27 lunar mansions, and yellow backgrounds for the Sun, Venus and Jupiter (diagram: Yukio Ôhashi).

(b) The 27 Indian lunar mansions (YÔ re-arranged the order in accordance with the usual order of Indian lunar mansions, and added Sanskrit names in brackets):

47. "Of *Athawane* the horse's head picture has six circles, and the *Rakain* country". (Aśvinī)

46. "*Pagan* country is governed by the old cock's figure". (Bharaṇī)

45. "*Kiatteka* has a fowl's picture, and six circles". (Kṛttikā)

44. "Of *Rohane* the snake's-head figure has ten circles". (Rohiṇī)

37. "*Mecatthe* has of an antelope's head the picture, three circles, and *Haynthawade* country". (Mṛgaśīras)

36. "Of *Adara Daway* is the country". (Ārdrā)

35. "The *Brahmen's Buchia* has a boat's picture, and the *Dagoun* country". (Punarvasu)

34. "*Buchia* the crab constellation ten circles has". (Puṣya)

57. "Of *Athaletha* the horse's-yard picture, four circles has, and the *Thattoun* country". (Āśleṣā)

55. "*Matha* has of a monkey the figure, four circles, and the *Baranathe* country". (Maghā)

54. "Of *Pyouppabaragounne* the cow's picture three circles has". (Pūrva-Phālgunī)

52. "Of *Uttarabaragounne* the bullock's picture two circles has". (Uttara-Phālgunī)

65. "*Hathadda* of an elephant's head the picture has, *Dhagnawade* is its country". (Hasta)

64. "Of *Zeittara* the tiger's picture, has one circle, and the *Wethale* country". (Citrā)

63. "Of *Thuade* a great snake's-head picture, has three circles, and the *Thayndua* country". (Svātī)

62. "*Wethaga* has of a buffalo's head the picture, and fourteen circles". (Viśākhā)

27. "Of *Anurada* the peacock's picture has fifteen circles, and the *Zedouttara* country". (Anurādhā)

26. "Of *Seitta* the goat's picture five circles has, *Zedouttara* is its country". (Jyēṣṭhā)

25. "Of *Mula* the cat's picture five circles has, *Peenzalareet* is its country". (Mūla)

24. "Of *Pyouppathan* the lion's picture two circles has, *Mouttama* country it governs". (Pūrva-Āṣāḍhā)

17. "Of *Uttara* the lion's picture two circles has, *Moranun* country governing". (Uttara-Āṣāḍhā)

16. "*Tharawun* constellation a hermit's picture three circles has". (Śravaṇa)

15. "Of *Danatheidha* the fisherman's picture four circles has". (Dhāniṣṭhā)

14. "*Thattapescia* with a leopard's picture six circles has". (Śatabhiṣaj)

5. "*Pyouppa parabaik* of a cow the picture has, and two circles, *Patanago* country it governs". (Pūrva-Bhādrapadā)

4. “*Uttara-parabaik* a cow’s figure has, and two circles, and the *Kappelawut* country”. (Uttara-Bhādrapadā)

3. “*Rewade* an alligator’s figure has, *Kutheinnaroun* country, and nine circles it has”. (Revatī)

(c) Other Burmese asterisms:

6. “A couch is *Sataga* constellation, four circles it has, and the *Kathee* country”.

7. “The *Pyathat* of twenty-four circles, is of *Kieen* country the constellation”.

8. “The duck constellation, five circles has, *Shan* is its country”.

9. “The *Kyabuayn aroo* leaf is the *Talain* country constellation, it has seven circles”.

12. “The table constellation four circles has, of the *Kiayn* country the constellations”.

13. “*Zain* constellation eleven circles has”.

18. “The *Pangiayn* mountain constellation four circles has, of *Rakain* country the constellation”.

19. “*Tareindane* constellation four circles has, of *Yoodaya* country the constellation”.

20. “A couch is *Pagan* constellation with four circles, of *Shethæk* country

21. “The cloud constellation has five circles, of *Thulabe* the constellation”.

28. “The fowl male of *Peenza* constellation circles fifty has, of *Sawa* country the constellation”.

29. “The fowl female of *Utta* constellation eight circles has, of *Uzaung* country the constellation”.

30. “Of an alligator the *a-me-kah-han* is the picture of *Uttara* constellation with eight circles, and the *Lahu* country”.

31. “The balance constellation”.

33. “The mountain constellation four circles has”.

39. “*Buchia* constellation has eight circles, and *Yun* country”.

40. “*Zaduka* constellation four circles has, in a pair of fetters, of *Giun* country the constellation”.

42. “The *Kyay* ship of twenty-eight circles”.

48. “*Pozoke* a constellation of eight circles belongs to the *Talain* country, like the *Hayntha* male and female”.

49. “*Putthata* constellation seven circles has, of the *Raneezee* tree the fruit”.

50. “*Aykatheitta* a constellation of four circles of *Kale* country the constellation, is like a basin”.

51. “*Tarouttara* constellation two circles has, and the *Taroup* country”.

58. “The flag is *Pathatta* constellation, six circles it has”.

59. “*Eessa* constellation six circles has, of *Momain* country the constellation”.

60. “Of *Akap*, a constellation of eight circles, *Daway* is the country”.

66. “*Kobiape* constellation with eleven circles has, the *Myamma* country”.

67. “A fowl’s foot is *Thareiddha*, a constellation of four circles, of *Laynzayn* country the constellation”.

68. “A boat’s ladder is *Tareiddha*, a constellation of six circles, of *Kula* country the constellation”.

(d) The Sun, Venus and Jupiter:

1. “Of *Sunday* the star”.

11. “The morning constellation one circle has, of *Dunwun* plant the fruit”.

38. “Of *Friday* the Star”.

53. “Of *Wednesday* the Star”.

There is also a paper by Khin Zaw (1937). Khin Zaw used a star map in the form of a parallelogram in the ‘Bernard Free Library’, and he identified 27 Indian lunar mansions and eight (out of the nine) inner northern Burmese asterisms with modern Western constellations. There were already studies of Indian lunar mansions in Indian Sanskrit texts, but there were some differences between India and Burma as Khin Zaw pointed out, and this paper is an important early contribution.

In Myanmar, there are some star maps, among the most famous of which are the three star maps (two circular and one rectangular) drawn on the ceilings of corridors at the Kyauktawgyi Pagoda. This Pagoda was built in 1847 by King Pagan of the Konbaung Dynasty, and is located at Amarapura (near Mandalay). Fig. 24.9 shows an external view of the Pagoda.

The Kyauktawgyi Pagoda star maps include several asterisms that are not found elsewhere. The existence of these star maps was known to researchers of Burmese Buddhism, and a photograph of one of the star maps was published by Ōno and Inoue in 1978. This inspired both authors to visit the Pagoda. The first author (YÔ) visited the Pagoda twice, the first time in 1984. At the time, the taxi driver who accompanied YÔ was very helpful, and his hobby happened to be photography. Luckily, he had the same camera, so YÔ sometimes borrowed his wide-angle lens for some of the photographs.

Subsequently, YÔ was informed that the Japanese astronomical historian Minewo Nishiyama was interested in the Burmese asterisms, so he presented his photographs to him and Nishiyama started studying the asterisms, with help from some specialists in the Burmese language. In 1997 he published a paper in English (Nishiyama, 1997) and other papers in Japanese. In his 1997 paper Nishiyama presented a list of 161 heavenly bodies on the star map in the southern corridor of the Pagoda. He tried to identify the Burmese asterisms with modern Western constellations, but many of the asterisms included in the 27 Indian lunar mansions and in the nine northern Burmese asterisms could not be identified. Later, in 1996, Mr Nishiyama visited the pagoda to see the star maps for himself and to take his own set of photographs. Then, after the November 2017 SEAN History and Heritage Working Group Meeting in Mandalay, the first author of this chapter (YÔ) was able to visit the Kyauktawgyi Pagoda again, after a break of 33 years, this time in the company of Visanu Euarchukiati (who also has chapters in this book). Generally, speaking, YÔ found the star maps were well preserved and looked almost the same as 33 years earlier, but there were some small areas with damage.



Fig. 24.9 The southern entrance to the Kyauktawgyi Pagoda at Amarapura (photograph: Yukio Ôhashi, 2017).

Figs. 24.10, 24.12, 24.14 and 24.16 are YÔ's photographs taken in 2017, while Figs. 24.11, 24.13, 24.15 and 24.17 are his tentative (partial) identifications of asterisms (with the 27 Indian lunar mansions numbered with red numerals, and the nine northern Burmese asterisms with blue Roman numerals). In the case of Star Map 'B' (in the northern corridor), they are all asterisms with Burmese captions. In the case of the Star Maps 'A' (in the southern corridor) and 'C' (in the western corridor), there were other asterisms with Burmese captions, and further research is needed. Some asterisms without captions were identified by their figure and position. The identification of 27 Indian lunar mansions is not necessarily the same as the usual Indian identifications, but is YÔ's tentative identification based on the Burmese star maps. He referred to the identifications in Khin Zaw (1937) and Nishiyama (1997), but his identifications are not necessarily the same as theirs. Regarding the Roman numerals of the nine northern Burmese asterisms, YÔ tentatively followed Nishiyama's Japanese listing which was previously circulated through the internet. YÔ first presented his own identifications in Ôhashi (2018).



Fig. 24.10 Star map 'A' in the southern corridor of the Kyauktawgyi Pagoda, Amarapura (photograph: Yukio Ôhashi, November 2017).

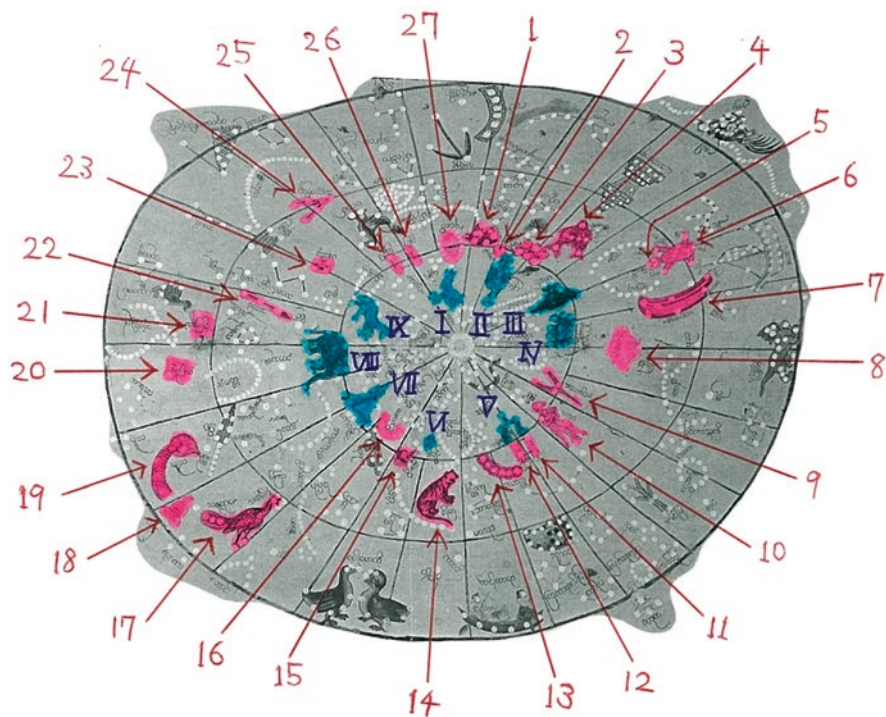


Fig. 24.11 YÔ's identification of the 27 lunar mansions and 9 asterisms in star map 'A'.



Fig. 24.12 Star map ‘B’ in the northern corridor of Kyauktawgyi Pagoda, Amarapura (photograph: Yukio Ôhashi, November 2017).

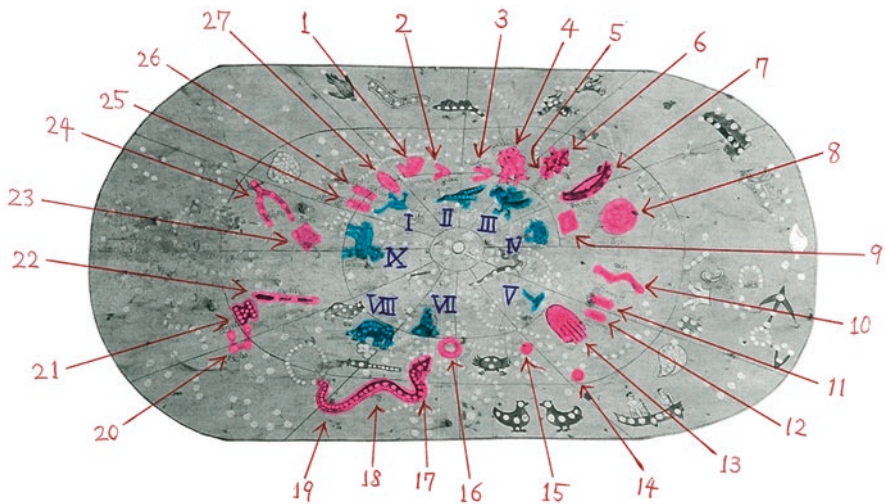


Fig. 24.13 YÔ’s identification of 27 lunar mansions and 9 constellations in star map ‘B’.



Fig. 24.14 Star map ‘C’ in the western corridor of Kyauktawgyi Pagoda, Amarapura (looking from its inner side) (photograph: Yukio Ôhashi, November 2017).

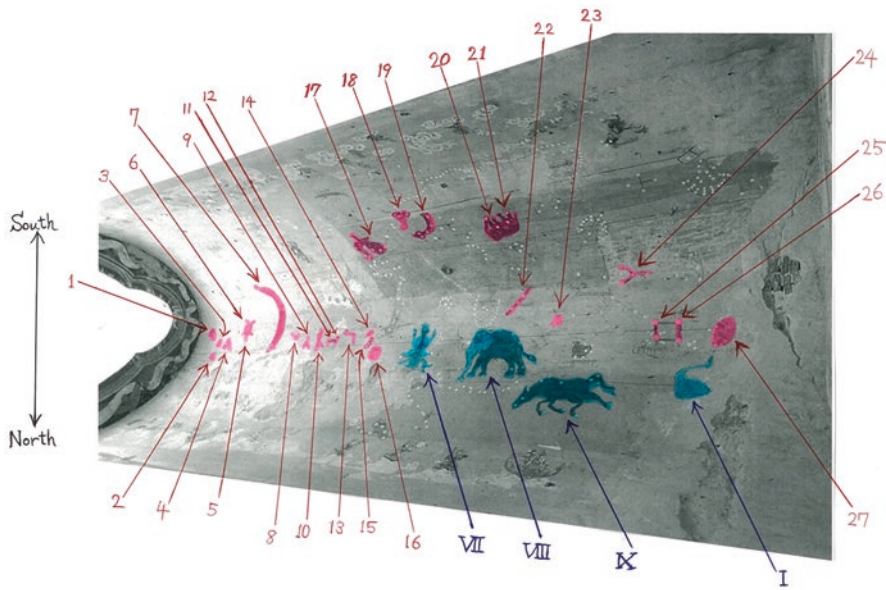


Fig. 24.15 YÔ’s identification of 27 lunar mansions and some asterisms in star map ‘C’ (looking from its inner side).



Fig. 24.16 The star map ‘C’ in the western corridor of Kyauktawgyi Pagoda, Amarapura (looking from its outer side) (photograph: Yukio Ôhashi, November 2017).

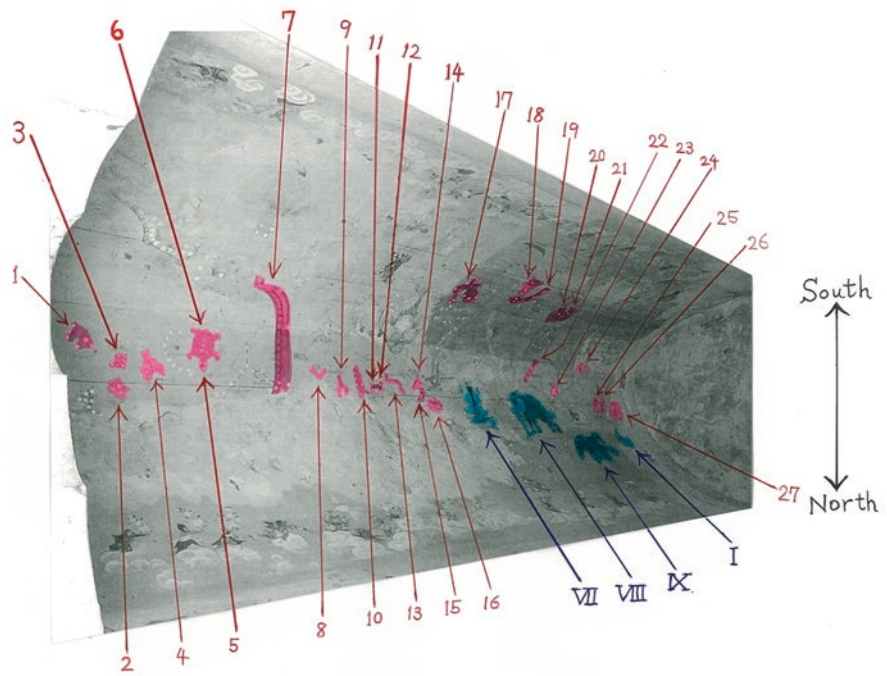


Fig. 24.17 YÔ’s identification of 27 lunar mansions and some constellations in star map ‘C’ (looking from its outer side).

Following are YÔ's identifications of the 27 lunar mansions and 9 northern asterisms shown in the Burmese star maps:

The 27 Indian lunar mansions:

1. *Aśvinī* (α , β , γ Ari)
2. *Bharaṇī* (35, 39, 41 Ari)
3. *Kṛttikā* (Pleiades)
4. *Rohiṇī* (Hyades)
5. *Mṛgaśīras* (λ , φ^1 , φ^2 Ori)
6. *Ādrā* (Orion)
7. *Punarvasu* (Gemini)
8. *Puṣya* (Cancer)
9. *Āśleṣā* (Western part (κ , λ , ϵ , μ) of Leo ?)
10. *Maghā* (α , η , γ , ζ Leo)
11. *Pūrva-Phālgunī* (δ , θ Leo)
12. *Uttara-Phālgunī* (β , 93 Leo)
13. *Hasta* (Northwestern part of Virgo ?)
14. *Citrā* (Spica)
15. *Svātī* (Arcturus)
16. *Viśākhā* (Corona Borealis)
17. *Anurādhā* (Northwestern part (β , δ , π) of Scorpius)
18. *Jyēṣṭhā* (α , σ , τ Sco)
19. *Mūla* (Southeastern part of Scorpius)
20. *Pūrva-Āṣāḍhā* (δ , ϵ (and γ , η ?) Sgr)
21. *Uttara-Āṣāḍhā* (ζ , σ (and τ , φ ?) Sgr) (* *Abhijit* (Disused in Burma))
22. *Śravaṇa* (α , β , γ Aql)
23. *Dhaniṣṭhā* (α , β , γ , δ Del)
24. *Śatabhiṣaj* (Western part of Aquarius)
25. *Pūrva-Bhādrapadā* (α , β Peg)
26. *Uttara-Bhādrapadā* (γ Peg, α And)
27. *Revatī* (Northeastern part of Pisces)

The 9 northern Burmese asterisms:

- I. *Byain* 'Heron', (Cassiopeia).
- II. *Kyi* 'Crow', (Perseus).
- III. *Hindha* 'Ruddy Sheldrake (duck)', (Auriga).
- IV. *Puzun* 'Crab', (head and forelimbs of Ursa Major).
- V. *Khyein* 'Balance (scales)', (hindlimbs of Ursa Major).
- VI. *Hsankyin* 'Hairpin', (Coma Berenices).
- VII. *Tanga* 'Fisherman', (Hercules).
- VIII. *Hsin* 'Elephant', (Cygnus).
- IX. *Myin* 'Horse', (Cepheus).

Since there are some small areas of damage on the innermost part of the star map 'C' when YÔ visited the Pagoda in 2017, he now presents some of his photographs taken in 1984 and 2017 as Fig. 24.18 and 24.20 (taken in 1984) and Fig. 24.19 and 24.21 (taken in 2017) for comparison.



Fig. 24.18 The outer part of star map 'C' (photograph: Yukio Ôhashi, 1984).



Fig. 24.19 The outer part of star map 'C' (photograph: Yukio Ôhashi, November 2017).



Fig. 24.20 The inner part of star map “C” (photograph: Yukio Ôhashi, 1984).



Fig. 24.21 The inner part of star map “C” (photograph: Yukio Ôhashi, November 2017).

24.4.1.3 Bugis Asterisms

There are several original asterisms in Indonesia. Ammarell (1999: 122–142) gives the following Bugis asterisms, found in southern Sulawesi:

- A. *bintoéng balué* ‘widow-before-marriage’, (α , β Cen).
- B. *bintoéng bola ké pang* ‘incomplete house’, ($\alpha \sim \delta$, μ Cru).
- B.1. *bembé’ é* ‘goat’, (Coal Sack nebula in Crux).
- C. *bintoéng bale mangngiweng* ‘shark’, (Scorpius (south)).
- D. *bintoéng lambarué* ‘ray fish, skate’, (Scorpius (north)).
- D.1. (identified without name) ‘lost Pleiad’, (Antares).
- E. *bintoéng kappala’ é* ‘ship’, ($\alpha \sim \eta$ UMa).
- F. *bintoéng kappala’ é* ‘ship’, ($\alpha \sim \eta$ UMa; β , γ UMi).
- G. *bintoéng balu Mandara’* ‘Mandar widow’, (α , β UMa).
- H. *bintoéng timoro’* ‘eastern star’, (Altair).
- J. *pajjékoé* (Makasar term) or *bintoéng rakkalaé* ‘plough stars’, ($\alpha \sim \eta$ Ori).
- J.1. *tanra tellu’é* ‘sign of three’, (δ , ϵ , ζ Ori).
- K. *worong-porongné* or *bintoéng pitu* ‘cluster or seven stars’, (Pleiades).
- M. *tanra Bajaoé* ‘sign of the Bajau’, (Magellanic Clouds).

Pelras (1987: 27–32) gives the following Bugis asterisms that relate to agriculture. The Roman capital letters indicate the corresponding asterisms in Ammarell’s list, while the corresponding Makassar names are shown in square brackets (the Bugis and Makassar languages are closely related).

1. (= K) *worong-mpolong* ‘Tuft’, [*borong-borong*], (Pleiades).
2. *wara-wara* ‘Burning Coal’, [*bara-bara*], (Aldebaran).
3. (= J.1) *tanra tellu* ‘Triple Beacon’, [*tanra tallu*], (δ , ϵ , ζ Ori).
4. *manu’* ‘Chicken’, [*jangang*], (Canopus, Sirius and Procyon).
5. *watang-mpata* ‘Job’s Tears Stalk’, [*batang-bata*], (α , β Leo).
6. (= B) *éppang* ‘Lame’, [*balla’ képpang*] ‘the Crooked House’, (Crux).
7. (= A) *walu* ‘Widow’, [*balu*], (α , β Cen).
8. (= D) *lambaru* ‘Rayfish’, [*lambaru*], (Scorpius (north)).
9. *tékkoroso* ‘Pushed Plough’, (Triangulum?).

Fig.24.22 is an overview of Bugis asterisms drawn by the first author of this chapter but based on Ammarell (1999) (indicated by Roman letters) and Pelras (1987) (indicated by numerals).

24.4.1.4 Philippines Asterisms

Ambrosio (2008a) lists the following Philippine asterisms:

- Batik* (Orion’s Belt)
- Mupu* (Pleiades)
- Bubu* (Big Dipper)
- Paliyama* (parts of Aquila)
- Mamahi Uttara* (North Star)

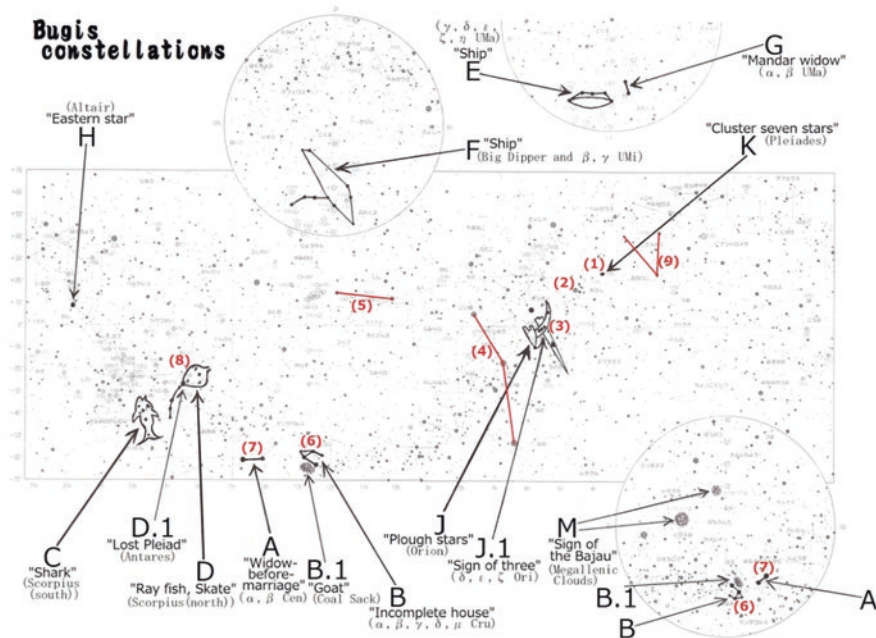


Fig. 24.22 Bugis asterisms, drawn by the first author of this chapter but based on Ammarell (1999) and Pelras (1987).

Saloka (Scorpius)

Anakdatu and *Sahapang* (Alpha and Beta Centauri)

Bunta (Southern Cross)

Lakag or *Maga* (morning star)

Mamahi Kagang,

Mamahi Pagi

Ambrosio (2008b) also mentions that *Balátik* (Orion) and *Moropóro* (Pleiades) were very important in the Philippines. As there are different names for asterisms in different languages in the Philippines, these names are different from those in the above list (see, also Ambrosio, 2010).

24.4.1.5 Additional Remarks

There are some similar asterisms in Southeast Asia. Fig. 24.23 shows Southeast Asian asterisms corresponding to Orion. In Fig. 24.23, A and B are from the Burmese star maps at the Kyauktawgyi Pagoda (Figs. 24.10 and 24.12) which depict a 'turtle'; C is the Thai asterism based on Saibejra (2012) which is a combination of the 'turtle' and 'plough', and D is the Javanese asterism based on Ammarell (1996) which is a 'plough'. By the way, in the Philippines, Orion is a 'trap' for hunting, as is shown on the front cover of Ambrosio (2010)—see Fig. 24.24).

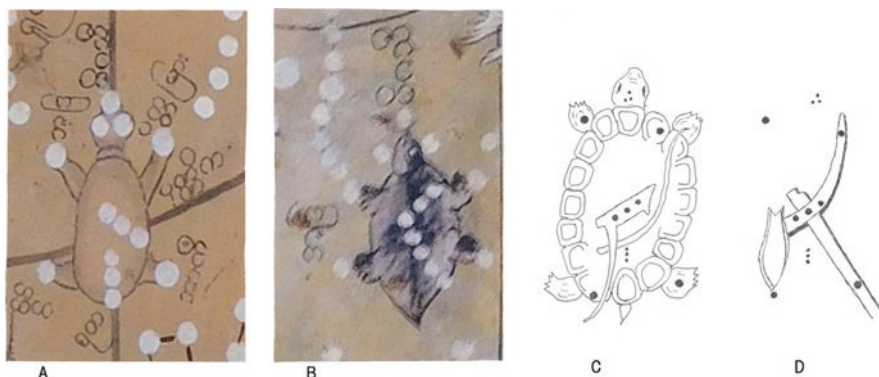


Fig. 24.23 Some similar of Southeast Asian constellations (diagram: Yukio Ôhashi).

Here are some recent works on folk-astronomy in Asia. Fig. 24.24 shows the front covers of Phengkaew (2009), Saibejra (2012) and Maison (2013) in Thai, Ambrosio (2010) in Filipino, and Kaifu (2014) in Japanese. It will be fruitful to discuss them at international seminars in the future.

24.4.2 *Timekeeping: The Pranotomongso Calendar of Java (YÔ)*

Various kinds of sun dials were used in several places. As they depend on local latitude, they are indigenous. We already saw some examples in the Geographical Overview section above. One interesting example is the gnomon that was used in Java in connection with the *Pranotomongso* calendar (see Fig. 24.4b). It was used to determine the months by observations of the noon shadow. The latitude of Java is around 7° S or so. The solar mid-day zenith distance varies from 30.5° N to 16.5° S, so, if we assume that the length of the vertical gnomon is 1, the length of the shadow (the tangent of the zenith distance of the Sun) varies from 0.589 south to 0.296 north. If the scale is divided into six equal segments, the lengths will be 0.589, 0.442, 0.294, 0.147 north, 0 (the root of the gnomon), and 0.148, 0.296 south. Then, considering the zenith to be 7° S, the Sun's zenith distance in each division will be approximately 23.5° , 16.9° , 9.4° , 1.4° , -7.0° , -15.4° and -23.5° . Then, using the formula

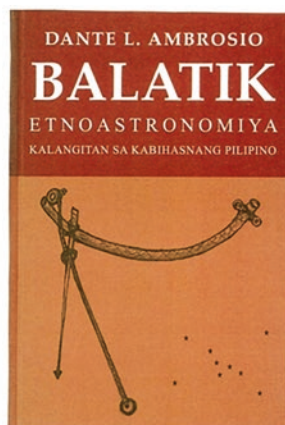
$$\sin \lambda = \sin \delta / \sin \varepsilon \quad (1)$$

where λ is the solar longitude, δ is the solar declination, and ε is the obliquity of the ecliptic, the longitudinal differences of each segment are approximately 43° , 23° , 21° , 21° , 24° and 48° . In the actual observation, the equation of the centre should

also be considered but this is disregarded. In the *Pranotomongso* calendar, one year has 12 months from the summer solstice, which consist of 41 days, 23 days, 24 days, 25 days, 27 days, 43 days, 43 days, 26~27 days, 25 days, 24 days, 23 days and 41 days respectively. This calendar must have been based on traditional astronomical knowledge that could only have been accumulated for a particular terrestrial latitude.



Phengkēw (2009)



Ambrosio (2010)



Kaifu (2014)



Saibejra (2012)



Maison (2013)

Fig. 24.24 Front covers of some recent books on the folk-astronomy in Asia (Ôhashi Collection).

24.5 Chinese Influences on Southeast Asian Astronomy (YÔ)

24.5.1 *The Twelve Animal Names of the Year*

The Chinese 12-year cycle of animals (the rat, ox, tiger, rabbit, dragon, snake, horse, goat, monkey, cock, dog and pig) is very popular in several parts of Asia, but with some local variations. Thus, in Vietnam the ox usually is replaced by the buffalo and the rabbit by the cat (see Fig. 24.25).

As there are Indian Jovian 12-year and 60-year cycles that are independent of the Chinese cycle, we should distinguish them carefully. For the Indian 12-year cycle in Burmese inscriptions see Furnivall (1922).

The Chinese 12-year animal cycle is widely used in Thailand. We have seen that it was used in the inscription of Ram Khamhaeng, along with the Indian Śaka Era. And also, the Chinese 60-year cycle (a combination of the 10-year cycle and the 12-year cycle) was used in the Vietnamese traditional calendar, and also by Thai people outside the Central region—for the case of the Thai calendar see Eade (1995: 24–25).

Fig. 24.25 is an example of the figures of the 12-year animal cycle used on modern Vietnamese postage stamps to celebrate traditional New Lunar Year's Day 'Tết', while Fig. 24.26 is an example of the figures of the 12-year animal cycle used on modern Thai postage stamps.

24.5.2 *The Chinese Luni-solar Calendar*

Serial numbers are used to denote lunar months in Thailand. De La Loubère (1693: 168–169) recorded that this method was used during the Ayutthaya Dynasty of Siam, and Eade (1995: 28–29) mentions some variations of this method in Thailand.

This method is not found in India, where the names of the lunar mansions are usually used to denote lunar-months. The first author of this chapter (YÔ) suspects that the idea of using serial numbers might be of Chinese origin.

In ancient China, the 19-year cycle was used for intercalation. This 19-year cycle was not popular in ancient India, but was very popular in ancient China. YÔ suspect that the 19-year cycle used in Southeast Asia derived from China (and see examples of what we call 'multiplex astronomy' in Subsection 24.9.3 below).

24.6 Indian Influences on Southeast Asian Astronomy (YÔ)

The initial influence of India on East and Southeast Asian astronomy occurred with the spread of the Hindu religion, followed soon after by Buddhism (Ôhashi, 2017a). Various aspects of this influence are discussed below.



Fig. 24.25 Figures of the 12-year animal cycle used on modern Vietnamese postage stamps (Ohashi Collection).

24.6.1 The Zodiac

The 12-sign zodiac originated in Mesopotamia, and was widely used throughout the Hellenistic World. Then it was introduced to India, the Islamic World, and the West.

The 12 zodiacal signs were introduced into India in around the third century CE from the Hellenistic World, and then introduced into Southeast Asia, as well as East Asia, from India. As the zodiac also is used in the Islamic World, the names of the signs of the zodiac have an Arabic origin in some areas of Southeast Asia, while the names derive from Sanskrit in other areas.

In India, the zodiacal signs were usually fixed to their sidereal positions around the second half of the sixth century CE. These Indianized zodiacal signs are used to regulate some Southeast Asian calendars. For example, the Thai festival of 'Songkrān' is held when the Sun enters Aries. As the signs are fixed to their sidereal positions, the Sun's entrance to the sign Aries, which corresponds to the vernal



Fig. 24.26 Figures of the 12-year animal cycle used on modern Thai postage stamps (Ôhashi Collection).

equinox, is now different from the usual (Western tropical) vernal equinox, due to precession. So ‘Songkrān’ is now celebrated on 13–15 April.

In present-day Thailand, the Sanskrit names of zodiacal signs are used for the names of the 12 months of the Gregorian calendar. Fig.24.27 is an example of the zodiacal signs from the front page of a modern Burmese calendar that the first author of this chapter purchased at a book shop in Mandalay at the time of the November 2016 Working Group meeting.

24.6.2 Lunar Mansions

India has 27 or 28 lunar mansions, and China has 28 lunar mansions. Originally, they probably were independent (see Chakravarty, 1987; Nakamura, 2017). Eade (1995: 31–37) discusses how the 27 lunar mansions were used in Thai and Burmese traditional calendars. In the Thai traditional calendar the position of the Moon is indicated by the Indian 27 lunar mansions as well as the 12 zodiacal signs. And as

we have seen, the 27 lunar mansions also are shown on the star maps at the Kyauktawgyi Pagoda in Amarapura, Myanmar (see Figs. 24.10, 24.12, 24.14 and 24.16).

24.6.3 *The Seven-day Week*

The 7-day week originated in the Hellenistic world, and was then introduced to India, the Islamic World, and the West. The 7-day week was introduced into India in around third century CE from the Hellenistic World, and then introduced into Southeast Asia and East Asia from India. In Thailand and Myanmar, the day of the week of one's birth is considered to be very important. As the 7-day week also is used in the Islamic calendar, the name of the days of the week originates with Arabic in some areas and from Sanskrit in other areas.

In Thailand, there are symbolic colours for the days of the week. Sunday is red, Monday is yellow, Tuesday is pink, Wednesday is green, Thursday is orange, Friday is blue, and Saturday is purple (Saibejra, 2012: 10; Segaller, 2005: 195). In Thai astrology, Wednesday is divided into day-time and night-time (after 6pm), and night-time corresponds to Rāhu. Sometimes, the symbolic colour of the night of Wednesday is considered to be black.

In Myanmar, Sunday is symbolized by the Garuda; Monday by the tiger; Tuesday by the lion; Wednesday (forenoon) by the elephant with tusks; Wednesday (afternoon) by the elephant without tusks; Thursday by the rat; Friday by the guinea-pig; and Saturday by the dragon. Sunday, Monday, Tuesday etc. usual relate to the Sun, Moon, Mars etc. In Myanmar, Wednesday (forenoon) usually relates to Mercury and Wednesday (afternoon) to Rāhu. Usually, the first letter of one's name is related to the day of the week of one's birth. Shway Yoe (1910: 4–6) and Meiji Soe (2012: 57–69) give detailed descriptions. Fig. 24.28 is an example of the figures of the days of the week on a modern Burmese calendar that the first author purchased in a book-shop in Mandalay at the time of the November 2016 SEAN History and Heritage Working Group Meeting.

24.6.4 *Indian Eras*

There are several eras in India. We have seen that the Indian Śaka Era was used in the inscriptions in Champa, Thailand (particularly that of King Ram Khamhaeng), Cambodia, Burma, Indonesia, etc. Also, the Buddhist Era is used in some regions of Mainland Southeast Asia. In Thailand, the Buddhist Era ('Phuttha-sakkarāt' in Thai) is usually used (Buddhist Era = CE + 543). It should be noted that the 0th year of the Buddhist Era begins from 544 BC in Thailand, Laos and Cambodia, while the 1st year of the Buddhist Era begins in 544 BC in Burma and Sri Lanka.

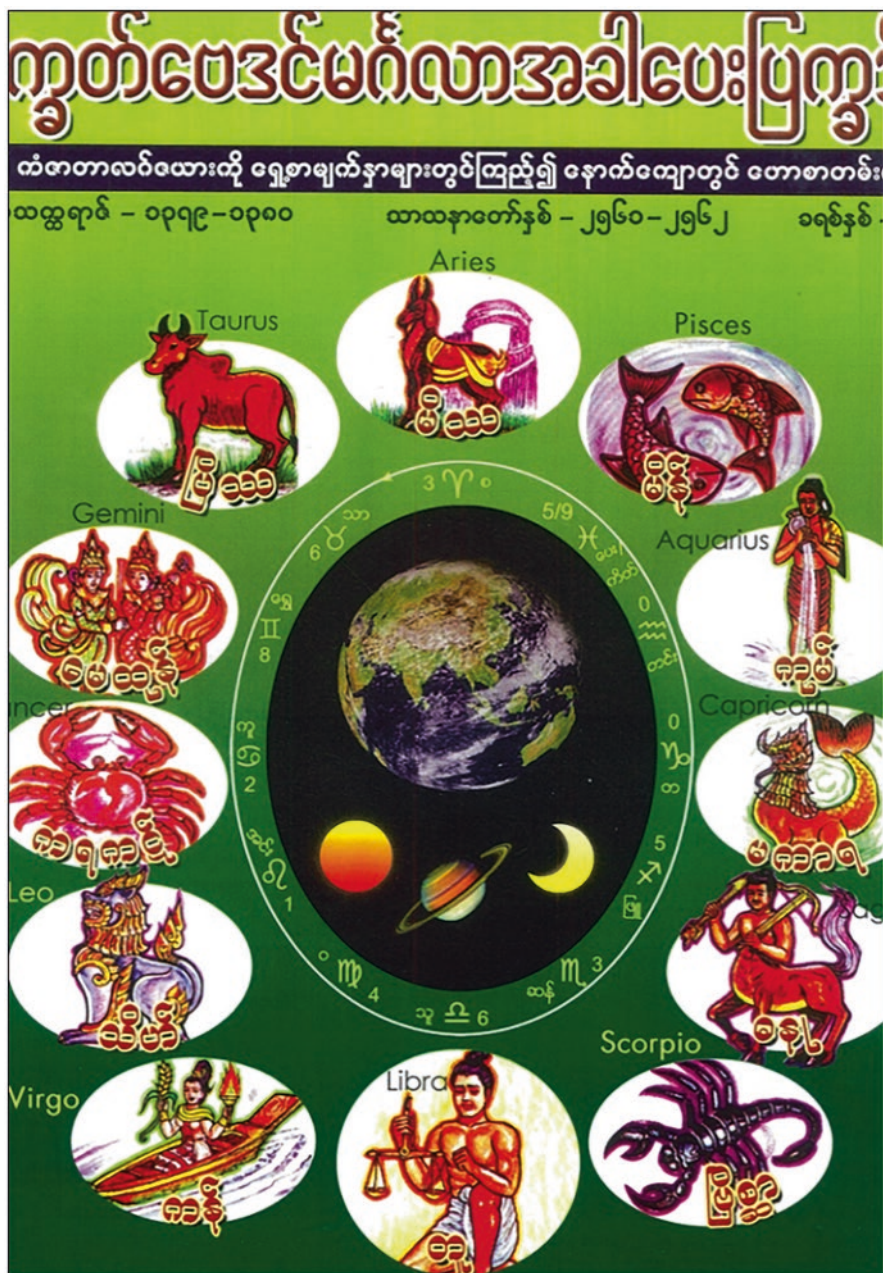


Fig. 24.27 Figures of zodiacal signs from the front page of a modern Burmese calendar (Ôhashi Collection).

24.6.5 *The Sidereal Year*

The traditional calendars of Mainland Southeast Asia (except for Vietnam) use the sidereal year, and this is of Indian origin. This is the result of the zodiacal signs being fixed in their sidereal positions around the second half of the sixth century CE, as we saw above in the section on the ‘zodiac’.

24.6.6 *The Indian Luni-Solar Calendar*

Hindu traditional calendars are mostly luni-solar calendars, and they were introduced into Southeast Asia. Since the Chinese traditional calendar is also a luni-solar calendar, it is sometimes difficult to distinguish Indian and Chinese influences in this respect.

One Indian specialty is the Hindu method of dividing one synodic month into a white half month (from New Moon to Full Moon) and a black half month (from Full Moon to New Moon). This method also is used in some Southeast Asian calendars. This division is absent in the Chinese luni-solar calendar. Another Indian specialty is to use the Sanskrit names of the lunar mansions as the names of the lunar months.

And also, one synodic month is divided into 30 *tithis* in the Hindu calendar. In the Hindu calendar, one *tithi* (or lunar day) is usual a period during which the longitudinal difference between the Sun and Moon changes by 12° . This kind of lunar day is also used in some Southeast Asian calendars.

The Indian Jovian 12-year cycle, which is independent of the Chinese 12-year cycle of animals, was used in Burmese inscriptions (Furnivall, 1922).

In India, there are several traditional astronomical calendars in different geographical areas and written in different languages. Fig. 24.29 shows some examples of Indian traditional astronomical calendars in the Hindi language, and Fig. 24.30 shows an example of a page in a Hindi traditional astronomical calendar.

Fig. 24.31 is an example of a traditional Thai astronomical calendar. Here, we can see that the positions of the Sun, Moon, planets and the ascending node of the lunar orbit at the end of the day (24:00) are given in sidereal signs, degrees and minutes. We also can see that the Sun enters sidereal Aries in 14 April 1999, which is the day of Songkrān. Ketu usually means the descending node of the lunar orbit or, sometimes, comets or meteors in Hindu astronomy, and usually means Neptune in Thai astrology (see Suriyā’ārak, 1983), but the position of Ketu in Fig. 24.31 is different from these, and its meaning is not clear. We can see that the Thai traditional luni-solar date has a maximum on the 15th, because one sidereal month is divided into two half months. This is Indian style.

24.6.7 Time-keeping

Wallace (1869: Chapter 28) recorded a water clock used on a local Makassan ship (Fig. 24.32, C is a drawing by the first author of this chapter based on Wallace's description). This floating type of water clock was used in India from the end of the fifth century or so, and was very popular (Fig. 24.32, A and B are photographs of the water clock in the Rao Madho Singh Museum, Kota, Rajasthan, India). For details of Indian water clocks see Ôhashi (1994).

24.6.8 Horoscopes

Horoscopes are found in inscriptions throughout Southeast Asia (e.g., for Thailand, see Eade, 1996). And also, there are several other examples that are popular till now (see Fig. 24.1, Fig. 24.2 and Fig. 24.32 in Thai and Burmese calendars).



Fig. 24.29 Some Indian traditional astronomical calendars in the Hindi language (Ôhashi Collection).

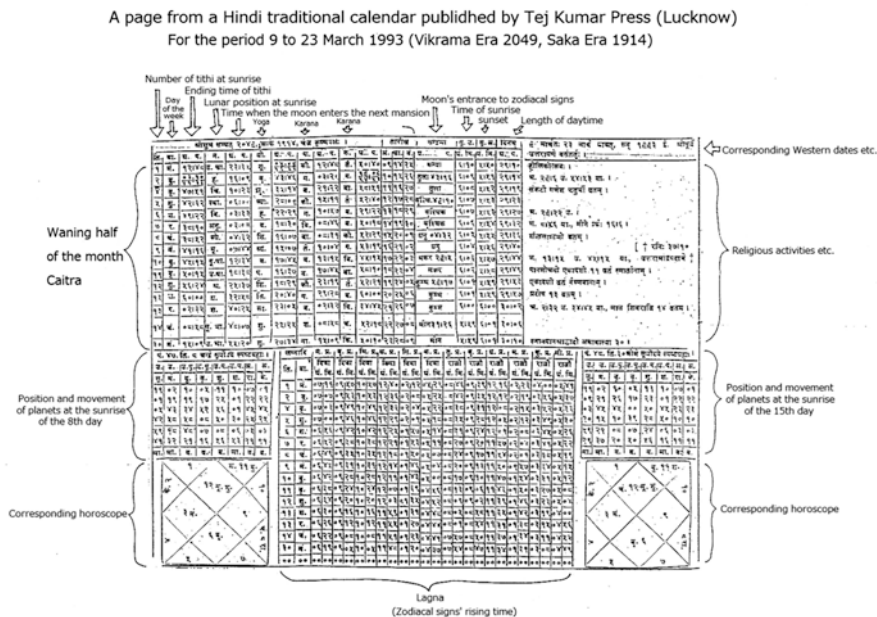


Fig. 24.30 A page from a Hindi traditional astronomical calendar, with comments by Yukio Ôhashi.

24.6.9 *Rāhu: The Demon Who Produces Eclipses*

In Indian astronomy, the ascending node and descending node are called Rāhu and Ketu. The legend of Rāhu is found widely throughout Southeast Asia (Fig. 24.33).

24.6.10 *Cosmology: Mount Meru, etc.*

The influence of Indian cosmology is found at several places in Southeast Asia (Quaritch Wales, 1977) and Swearer (2010). Bogle (2016) explains the Buddhist cosmology in Burma with several beautiful figures. Buddhist cosmology (the Mount Meru model) is mainly based on the *Abhidharma-kośa* of Vasubandhu (ca. fourth–fifth centuries CE). There are several traditional pictures of Mount Meru in Thailand. And also, it is said that the central tower at the Angkor Wat temple complex in Cambodia symbolizes Mount Meru.

King Lithai of the Sukhothai Dynasty of Thailand wrote the *Trai-phūm* (Three Worlds) in the mid-fourteenth century (Reynolds and Reynolds, 1982). This is a celebrated text of traditional Thai cosmology, and some information about traditional astronomy is also found there.

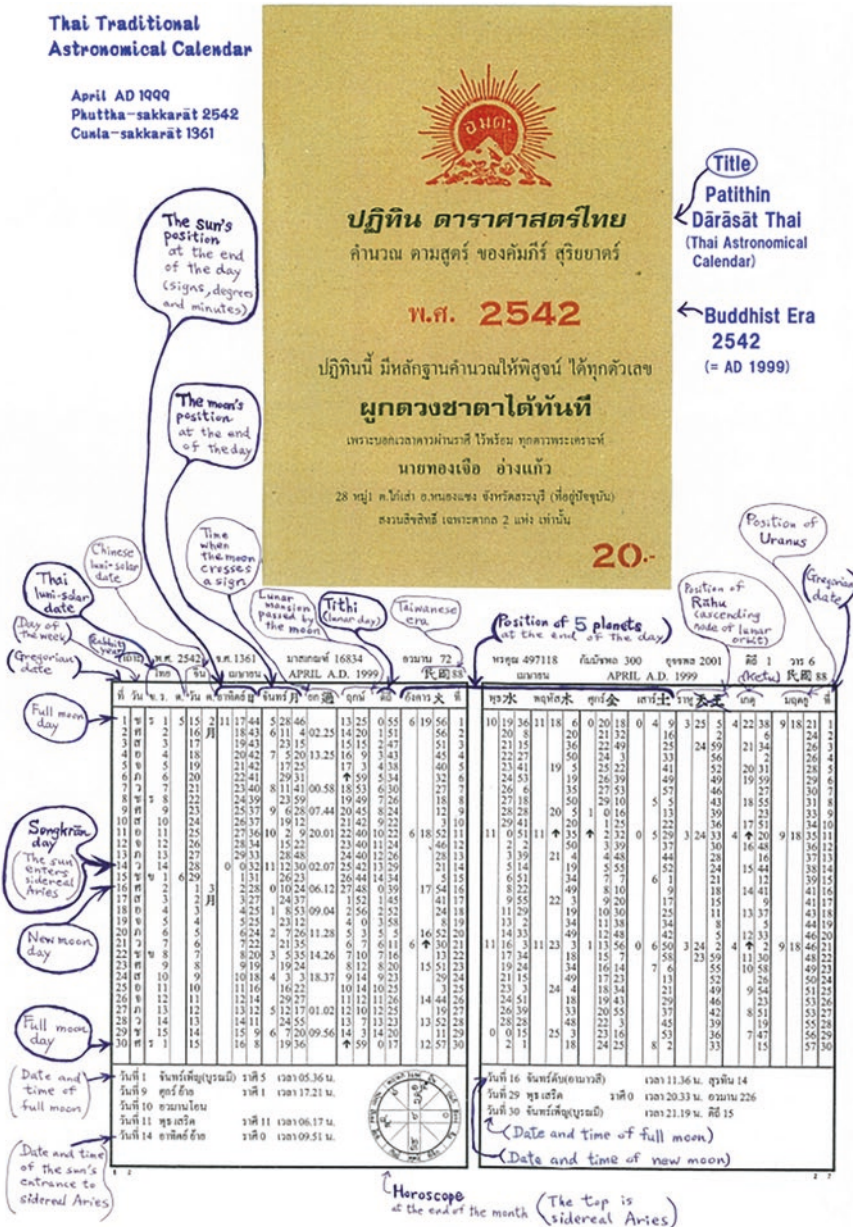


Fig. 24.31 An example of a Thai traditional astronomical calendar, with comments by Yukio Ôhashi (Ôhashi Collection).

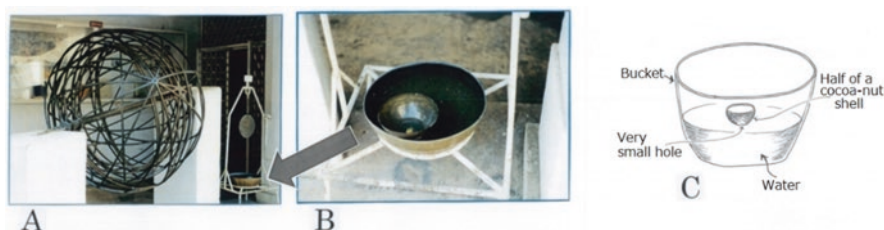


Fig. 24.32 Indian water clock (A and B) and a drawing (C) of a Makassan water clock (photographs and drawing: Yukio Ôhashi).

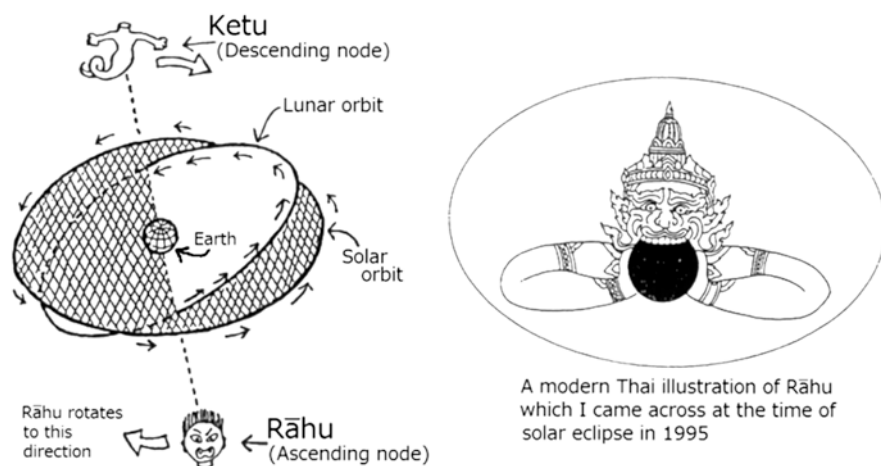


Fig. 24.33 Rāhu and Ketu (original artwork: Yukio Ôhashi).

24.6.11 Architecture

The Indian traditional science of architecture is called ‘Vāstu-sāstra’ in Sanskrit. For the determination of directions, astronomical methods were used. An interesting paper about the application of this principle in the planning of Chiang Mai city in Thailand was presented at the February 2020 H&H Meeting (see Saelee et al., [2021](#)).

24.7 The Influence of Islam (YÔ)

Islam spread through island Southeast Asian between the thirteenth and sixteenth centuries, and quickly became the primary religion of what are now known as Malaysia and Indonesia. The astronomical impact of this newly-introduced religion is discussed below.

24.7.1 *The Islamic Lunar Calendar*

Basically, when the new crescent Moon is observed in the western sky in the evening, a new Islamic lunar month begins from sunset. So, the length of a month (29.530589 days) is exact. One year is 12 lunar months regardless of the season.

There are some simplified methods to predict lunar visibility. Proudfoot (2006) discussed the methods used in early island Southeast Asia. One method is trigesimal calendars that use a 30-year cycle (360 months), which has 11 leap years. This means that one lunar month is $(30 \times 6 \times (29 + 30) + 11) / (30 \times 12) = 29.530556$ days. A simpler method is octaval calendars that use an 8-year cycle (96 months), which has 3 leap years. It means that one lunar month is $(8 \times 6 \times (29 + 30) + 3) / (8 \times 12) = 29.53125$ days. Proudfoot says that the octaval calendar was widely used in pre-nineteenth century Southeast Asia. And we saw in the Geographical Overview Section that there is also a 120-year cycle (1440 months) with 44 leap years, which consists of 42,524 days. This is actually just four times the 30-year cycle.

At present, the visibility of the crescent Moon is still very important (for the case of India, see Ôhashi, 2006b), and there are several reports about it. In Japan, meanwhile, Kawada (2008) reports that there is a ‘Ruyat-e-Hilal Committee-Japan’ that observes the crescent Moon at the beginning and the end of the month Ramadan (the month of fasting), and if it is not observed then they follow the decision of the Malaysians. However, there are some Muslims in Japan who follow Turkey’s decision. So, sometimes there may be a difference of one day in different Islamic calendars from the same area.

According to Horii (1987) who studied in Malaysia, some people prefer to make actual observations while others prefer calculations, so sometimes different states of Malaysia use slightly different Islamic dates.

24.7.2 *Time-keeping and Qibla*

There are five prayers called ‘*ṣalāt*’ daily, and time-keeping is very important for Muslims. There was an interesting record of the prayer time in the Dutch East Indies (now Indonesia) in the early twentieth century (Izutsu, 1942: 29–30), as follows:

- (1) *ẓuhr* (after the time when the Sun is going to move from its highest point and before the shadow becomes the same length (or double length) as the height of itself): starting from 12 noon in the Dutch East Indies (while starting shortly after 12 am in Mecca).
- (2) ‘*aṣr* (in the afternoon after *ẓuhr* and before the Sun is going to set): done around half past 4 pm in the Dutch East Indies (while done about 3 hours after *ẓuhr* in Mecca).
- (3) *maghrib* (after complete sunset within evening twilight): done around 6 pm in the Dutch East Indies (while done right after sunset in Mecca).

- (4) *ishā* (at night after evening twilight and before morning twilight (before sleeping): done around half past 8 pm in the Dutch East Indies (while done about two hours after *maghrib* in Mecca).
- (5) *ṣubḥ* (after *ishā* and before sunrise): done between 4 am and 5 am in the Dutch East Indies (while done about one and a half hour before sunrise in Mecca).

The method of determining time in the pre-modern Southeast Asian Islamic world needs to be studied further.

The direction of the Kaaba in Mecca is called ‘Qibla’, and prayers should be done in this direction. According to Hurgronje (1906), Acehnese people and Javanese people considered that the rising three stars of Orion indicate Qibla. Ammarell and Tsing (2015) say that Banjar wet-rice farmers (in South Kalimantan) also saw the three stars of Orion’s Belt as pointers to Qibla.

Lunar visibility, prayer times and the direction of Qibla are all very important in Islamic daily life, and there have been several attempts to modernize them. One attempt was made by the well-known Malaysian astronomer Mohammad Ilyas (1984, 1997).

24.8 Western Astronomers and Astronomy (YÔ and WO)

From the mid-sixteenth century Jesuit missionaries visited Southeast Asia, and Western astronomy was introduced into Asia through them. For example, on CE 11 December 1685 King Narai of Siam observed a lunar eclipse with Jesuits sent by the French King Louis XIV (Gislén et al., 2018; Orchiston et al., 2016). This event is depicted in Fig. 24.34. Later, Siam’s King Rama IV studied Western astronomy as well as traditional Siamese and Mon astronomy, and he predicted and observed the total solar eclipse of AD 1868 (Soonthornthum and Orchiston, 2021) from Wah-koa, a coastal locality in southern Siam.

It should be noted that NARIT (the National Astronomical Research Institute of Thailand) has published the following series of books on the history of modern astronomy in Siam/Thailand in the Thai language: the reign of King Narai of the Ayutthaya Dynasty by Phumathon (2012a); the reign of King Mongkut (Rama IV) by Phumathon (2012b); the reigns from King Rama V to King Rama VIII by Euarchukiati (2015); and the reign of King Bhumibol Adulyadej (Rama IX) by Soonthornthum and Thajchayapong (2012). Meanwhile, for English-language overviews of the history of Thai astronomy see Soonthornthum (2017) and Orchiston et al. (2019).

During the nineteenth and twentieth centuries, Southeast Asia also was host to Western expeditions intent on observing the total solar eclipses of 1868 (Mumpuni et al., 2017; Orchiston and Orchiston, 2017a; Orchiston et al., 2021c); 1871 (Mumpuni et al., 2017); 1875 (Euarchukiati, 2021; Kramer and Kramer, 2017); 1901 (Pearson and Orchiston, 2017); and 1929 (Noor and Orchiston, 2021; Soonthornthum et al., 2021). Fig. 24.35 shows the eclipse camp at Wah-koa, Siam, where a French expedition successfully observed the total solar eclipse of 18 August

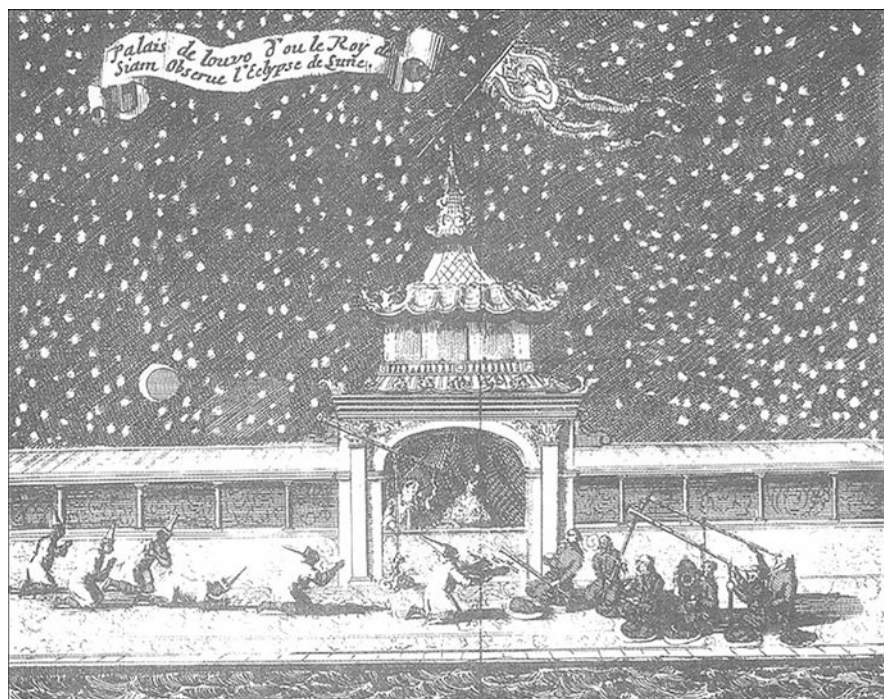


Fig. 24.34 A drawing supposedly showing King Narai (in the pavilion) and French Jesuit astronomers using telescopes to observe the 11 December 1685 lunar eclipse from the King's country retreat near Lop Buri, but as Orchiston et al. (2016) have shown, this contains considerable artistic license (after Tachard, 1686).

1868 at the invitation of King Rama IV (Orchiston and Orchiston, 2017a). King Rama IV's own eclipse camp was located nearby.

We also should consider Western influence on the traditional astronomy of Asia. For example, the last Chinese traditional calendar, the *Shixian-li* (時憲曆) (which was used in China from CE 1645) was based on Western astronomy. Subsequently, the *Shixian-li* calendar was adopted in Vietnam in CE 1813. In Vietnam, in CE 1837 the prime meridian was declared to pass through its capital Hue, and the longitudinal difference between Beijing and Hue was determined in 1841. At that time, longitude was measured from Paris. Here, also, we can see Western influence on the traditional astronomy of Vietnam (cf. Phâm and Lê, 2021).

Finally, the history of recent astronomy in some areas of Southeast Asia after the introduction of Western astronomy has been recorded in various chapters in the books edited by Batten (2001), Hearnshaw and Martinez (2007) and Nakamura and Orchiston (2017).



Fig. 24.35 A photograph of the 1868 French eclipse camp at Wah-koa, Siam, showing instrument huts and the 40-cm (left) and 20-cm (right) reflecting telescopes set up outdoors (courtesy: Archives, Observatoire de Marseille, 132 J 84).

24.9 Expanding on the Concept of ‘Multiplex Astronomy’

24.9.1 *Introduction: Changing Astronomical Systems (WO)*

Near the start of this chapter we introduced the concept of ‘multiplex astronomy’, where there is evidence that the original local astronomical knowledge system has been modified by the later superimposition of other astronomical influences from China, India, the Islamic World, the West, or any combination of these. In mainland Southeast Asia the oldest known astronomical systems usually date to the local arrival of farming communities, and given this comparatively short time frame it is relatively easy to identify the external astronomical influences, as we have seen in Sections 24.4–24.7 above.

However, the situation in island Southeast Asia and the Malay Peninsula tends to be much more complicated, given that the existing negrito populations of southern Thailand, the Malay Peninsula and various islands in the Philippines archipelago possess astronomical knowledge bases that evolved from the ancestral astronomical system that was introduced 60,000–70,000 years ago when the first *Homo sapiens* settled this region.

Then around 4000 years ago Austronesian populations from Taiwan spread throughout much of island Southeast Asia, imposing their language and their culture—including their astronomical beliefs and practices—on the existing negrito (Australo-Melanesian) populations, before many negrito groups withdrew, or were

driven, into the heavily-forested mountainous interiors of the islands in the Philippines archipelago, where they could maintain their identities in cultural and genetic isolation.

Around 3000 years ago Austro-Asiatic populations from South China and Vietnam settled the Malay Peninsula and southern parts of island Southeast Asia, bringing their own distinctive astronomical knowledge systems, which were closely linked to rice-cultivation. They in turn imposed their astronomical beliefs and practices on the existing negrito populations of southern Thailand and Malaysia, and on the Austronesians. Meanwhile, their own ecologically-related astronomical beliefs and practices soon evolved as they adjusted to successfully growing rice in localised tropical environments dominated by monsoons. It is also possible that some Austro-Asiatic groups chose to adopt various astronomical beliefs of the negritos and Austronesians, and incorporate them into their own astronomical knowledge systems.

Later, the Hindu religion and Islam spread throughout much of island Southeast Asia, introducing their own distinctive astronomical concepts and practices, and leaving their imprints of the daily lives of the local people. Then within the past 500 years the Malay Peninsula and all of the islands of Southeast Asia have experienced colonialism, with each of the European occupying nations (Great Britain, the Netherlands, Portugal, Spain and the United States of America) having its own established reputation in scientific astronomy and—in more recent years—in astrophysics.

Finally, we should note that in addition to the astronomical beliefs and practices introduced over the millennia to the Malay Peninsula and island Southeast Asia by different ethnic groups, religions and perhaps colonialism, astronomical knowledge systems also were liable to undergo change through cultural evolution and adaptation (cf. Orchiston and Orchiston, 2017b), and in response to major environmental changes—and especially those that occurred in sea level, temperature and rainfall over the last 20,000 years (following the last phase of glaciation).

When viewed from a chronological perspective, we can see that the Malay Peninsula and island Southeast Asia experienced a succession of ‘notable events’, each of which had the potential to impact significantly on existing astronomical systems. In chronological order, these events were:

- (1) Occupation of the region by Australo-Melanesians 70,000–60,000 years ago
- (2) Rapid and sustained post-glacial sea level rise between 22,000 and 6000 years ago
- (3) Occupation of the region by Austronesians, beginning about 4000 years ago
- (4) Occupation of part of the region by Austro-Asiatics, beginning about 3000 years ago
- (5) The spread of Hindu religion throughout much of the region, beginning about 1200 years ago
- (6) The spread of Islam throughout much of the region, beginning about 800 years ago
- (7) The spread of Western colonialism throughout the region, beginning about 500 years ago

Let us now examine this interplay of changes in human populations, the environment, religion and colonialism in the Malay Peninsula and island Southeast Asia and their likely impacts on local astronomical systems. Since topics (5) and (6) and

have already been discussed earlier in this chapter, in Sections 4.6 and 4.7, they will only be examined briefly here. Therefore, our emphasis will be on topics (1) through (4) and (7). Note that all of these have also been discussed elsewhere in this book, particularly in Chapters 2 and 19.

24.9.2 *The Potential Impact of Environmental Change, Migrations, Religions and Colonialism on Island SE Asian Astronomical Systems (WO)*

24.9.2.1 *Australo-Melanesian Occupation of the Southeast Asian Region and Emerging Astronomical Systems*

The earliest inhabitants of the Southeast Asian region were hominids belonging to *Homo erectus* (e.g. see Sémah, et al., 2000; Zaim, et al., 2011) and later *Homo Floresiensis* (Brumm, et al., 2010; Sutikna et al., 2016), and we can assume that some of these ancestral populations would have been encountered when the first anatomically modern humans, belonging to the species *Homo sapiens*, migrated into the region between 70,000 and 60,000 years ago (see Dennell, et al., 2014; Swisher III et al., 1996). They originated in northern Africa, and it is thought they probably followed the coast on their journey eastward. Known now as Australo-Melanesians, they settled Southeast Asia and also Australia and New Guinea (O’Connell et al., 2018). Although sea levels (Voris, 2001) during the period 70,000–60,000 BP¹ were significantly lower than those of today (see Fig. 24.36), these newcomers still had to successfully negotiate challenging water crossings in order to reach the Philippines region (Heaney, 1985), and Sahul—the enlarged Australia-New Guinea continent (e.g., see Kealy et al., 2016, 2017; O’Connor, 2015; O’Connor et al., 2017).

We see evidence of the Australo-Melanesians today in archaeological sites and human skeletal remains found throughout Sahul, Sundaland (the enlarged SE Asian land mass) and the Philippines (see Fig. 24.36) and, indeed, the earliest dated evidence of this population influx is from Callao Cave in the northern Philippines, where human skeletal remains have been dated to 67,000 BP (Mijares et al., 2010). Other early human skeletal remains have been recovered from Tabon Cave on the island of Palawan (Détroit et al., 2004; Dizon, 2003; Dizon et al., 2002; Fox, 1970),

¹ In this section the term BP refers to ‘before the present’, where ‘present’ by international definition is the year 1950. Geological time, for the period we discuss in this Sub-Section, is covered by two epochs, the Pleistocene (from 2.58 million years BP to 11,700 BP) and the Holocene (from 11,700 BP to the present day). The Pleistocene has Early, Middle and Late stages, with the Late Pleistocene from 126,000 to 11,700 BP (Gibbard et al., 2010). Sometimes it is convenient to express these dates as 126 kyr BP and 11.7 kyr BP, while the Pleistocene began at 2.58 myr BP. The Pleistocene was characterized by continuous glacial and interglacial phases, but only the Late Pleistocene is relevant here, and most of our discussion in this Sub-Section will focus on the Holocene.

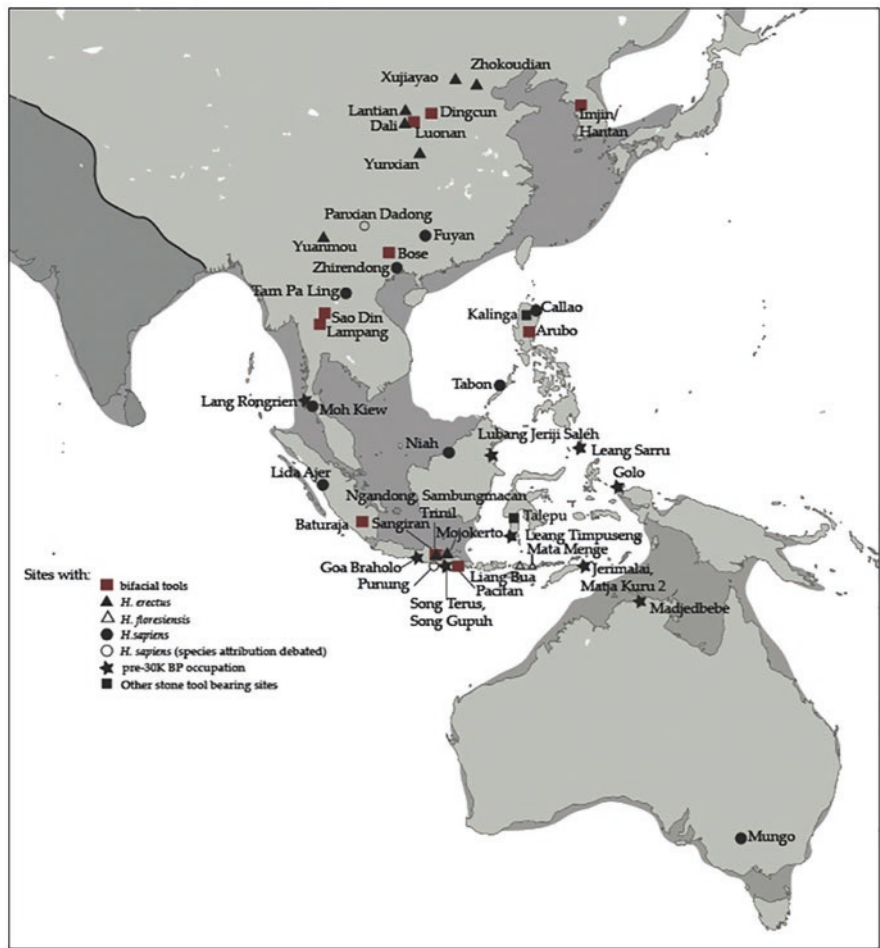


Fig. 24.36 A map showing East and Southeast Asian sites with bifacial tools, and human skeletal remains; the dark shading shows the land areas of ‘Sundaland’ (an enlarged mainland Southeast Asia) and ‘Sahul’ (conjoined Australia and New Guinea) when the seas were 110 m below current levels (after Roberts and Amano, 2019: Fig. 1).

and Niah Cave (Barker, et al., 2013; Curnoe et al., 2019; Hunt and Barker, 2014) in what is now northern Borneo.

The other even more conspicuous evidence of the Australo-Melanesian influx is the current existence of ‘negrito’ populations on the Andaman Islands in the eastern Indian Ocean (e.g. see Radcliffe-Brown, 1928), in southern Thailand, throughout the mountainous forested interior of the Malay Peninsula (Carey, 1976), and on various islands in the Philippines archipelago (e.g. Griffin and Estioko-Griffin, 1985; Rahmann and Maceda, 1955; Reid, 2013; Samson and Vitin, 2019). Their geographical distribution is shown in Fig. 24.37). The negritos are distinguished

from other indigenous ethnic groups by their short stature, dark skin and dark hair, but in Malaysia and the Philippines in particular they display considerable phenotypical heterogeneity. This is supported by genetic evidence. Studies using mtDNA (inherited through females) and Y chromosomes (inherited through males) show that as a group, the Philippines negritos can be clearly distinguished from the Malaysian negrito populations, which also differ significantly from the Andaman negritos (e.g. see Fig. 24.38). This would suggest that there were two or more different waves of Australo-Melanesians who settled mainland and island Southeast Asia (cf. Corny et al., 2017). Be that as it may, what these negrito groups offer us is an opportunity to possibly identify elements of the original Australo-Melanesian astronomical systems that were introduced to Southeast Asian with the first *Homo sapiens*. Studies conducted among the Aboriginal Australians (e.g. Hamacher and Norris, 2009); Nunn and Reid, 2016) show that astronomical and geomythological information can successfully be passed down from generation to generation for tens of thousands of years.



Fig. 24.37 The current geographical distribution of negrito ethnic groups in the Southeast Asian region; the larger circles in Malaysia and the northern Philippines indicate the presence of many negrito groups in these two areas (map: Wayne Orchiston).

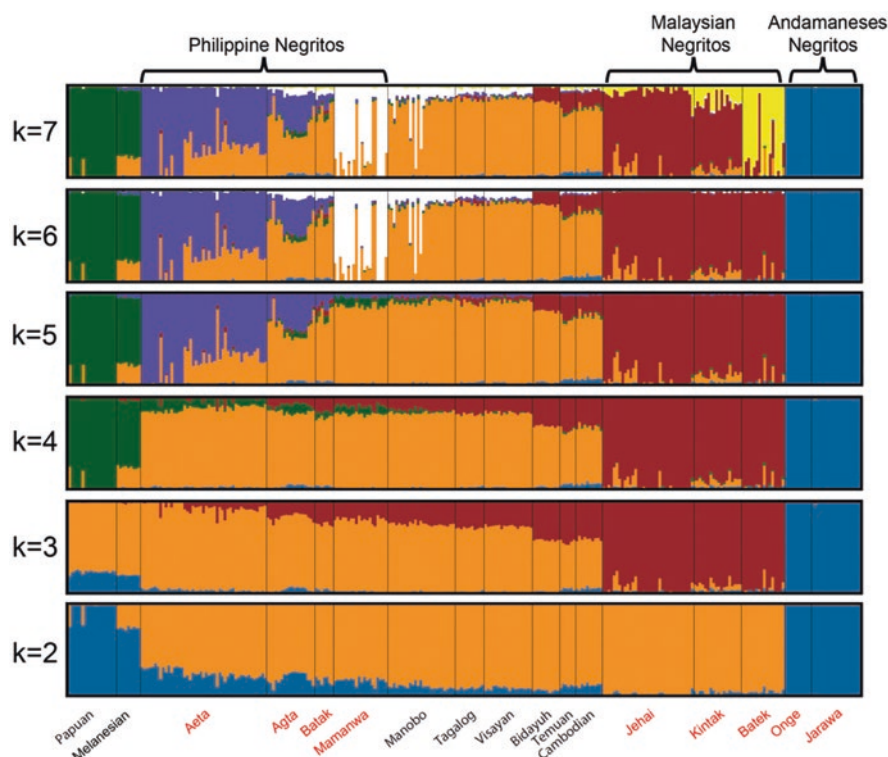


Fig. 24.38 A diagram showing the distinctive genetic signatures of the Andaman, Malay Peninsula and Philippine negrito groups (after Jinam et al., 2017: 2017).

Macauley et al. (2005) and Oppenheimer (2012) have both reviewed the genetic evidence, and concluded that anatomically modern man, *Homo sapiens*, exited East Africa on a single occasion, and took a route that crossed the entrance to the Red Sea, and ran along the coast of the Arabian Peninsula, round the coast of the Indian Subcontinent, into mainland and then island Southeast Asia. However, Petraglia et al. (2010: 305) believe this is an over-simplification:

Based on our review of the fossil, genetic, and archaeological evidence, we argue that the expansion of *H. sapiens* out of Africa was a complex process, and not simply the consequence of a single, rapid coastal expansion at 60 ka ago. We argue that that modern human expansion into Arabia is probably coincident with the ages of 130–70 ka for *H. sapiens* in the Levant, with populations reaching South Asia by 78 ka, and perhaps earlier.

These new inhabitants of Southeast Asia must have brought with them ancestral astronomical systems that formed part and parcel of their evolving cultural complexes. These Australo-Melanesians were hunter-gatherers and if later Aboriginal Australian ethnographic analogies are any indication we can assume that there were close correlations between human ecology and astronomy—so what was of prime importance here on the Earth often was projected into the sky and associated with

specific stars or asterisms, or even with dark lanes in the Milky Way (e.g. see Clarke, 2014, 2015; Fuller et al., 2014; Gullberg et al., 2020; Hamacher, 2015).

At this time sea levels were lower than today (see Fig. 24.39) and the Malay Peninsula, Sumatra, Borneo and Java all formed part of an extended SE Asian continent, with the area referred to as Sundaland. This is indicated by the dark grey areas in Fig. 24.36, but this map is to some extent misleading because the position of the coastline was constantly changing, sometimes even on a decade-by-decade basis—therefore within the span of a normal human lifetime—as sea levels oscillated between 70 kyr and 40 kyr. This obviously impacted on coastal exploration of the region, as did the presence of dense tropical rainforest, though Bird et al. (2005) and others have suggested that an E-W ‘savannah corridor’ across Sundaland helped facilitate the dispersal of populations throughout Sundaland, to the Philippines, islands in the Lesser Sunda chain in eastern Indonesia, and also to Sahul (the Greater Australian continent), all of which required challenging marine crossings (O’Connell et al., 2010). Yet archaeological, genetic and human skeletal evidence indicates that all of these water crossings were successfully negotiated (e.g. see Arenas et al., 2020; Gomes et al., 2015; O’Connell et al., 2018; Wurster and Bird, 2014).

As hunter-gatherers the Australo-Melanesians preferred the biotic diversity of coastal habitats, but they quickly adapted to the full range of environments encountered in Sundaland (Boivin et al., 2013). Roberts and Stewart (2018) introduce the term ‘adaptive plasticity’, and elaborate:

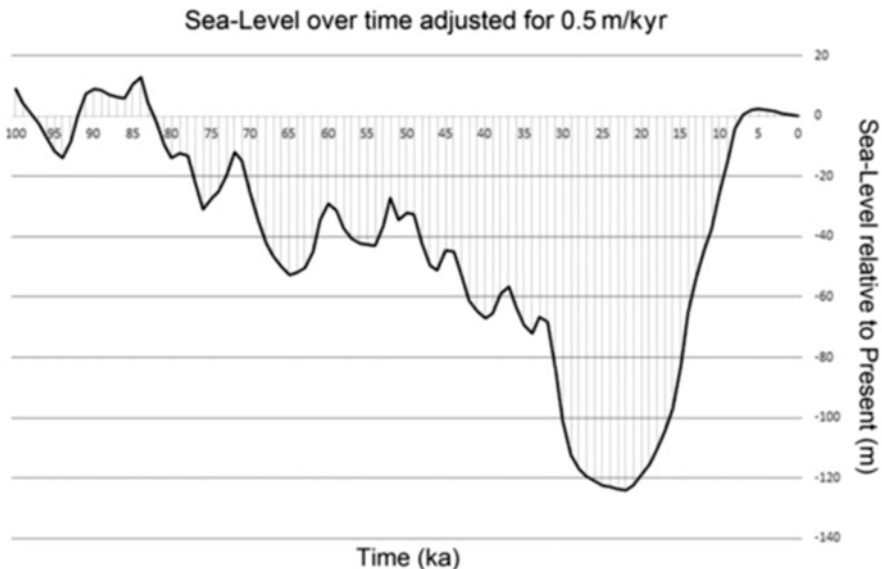


Fig. 24.39 A plot showing the changing sea levels in the Southeast Asian region over the past 100,000 years (after Lambeck and Chappell, 2001, but incorporating an isostatic uplift of 0.5 meters per thousand years, as calculated by Kealy et al. 2017).

... our species also documents a series of regional specializations in its adaptation to different rainforest environments. For example, at the Niah Caves, Borneo, humans deliberately maintained a mixture of grassland, peat swamp rainforest and evergreen rainforest to exploit primates, migratory suids and a variety of plant products. (*ibid.*)

Barker et al. (2007, 2013, 2020) document the Niah Cave evidence, while Morley (2016) notes that "... growing evidence shows that tropical rainforests, far from being unfavourable for human exploitation as was traditionally thought (see e.g. Bailey et al., 1989), have been successfully exploited for at least the past ~45 ka ..." Roberts and Petraglia (2015) review the accumulated evidence for Southeast Asia, and support this conclusion.

It was not just a matter of exploring and adapting to a new environment but the Australo-Melanesians had to adjust to an environment undergoing constant transformation. The sea level fluctuations documented in Fig. 24.39 provided an ever-changing landscape as rising and falling sea levels first drowned coastal habitats and then exposed them, leading to ecological regimes that slowly had to be altered over the course of a century or so. But superimposed on these coastal changes were climatic variations (see Bird et al., 2007; Lewis et al., 2008a; Westaway et al., 2007), which—as with sea levels—were intimately linked to glacial and interglacial phases. Variations in temperature and rainfall (e.g. see Fig. 24.40) also led to expanding and contracting biotic zones (and especially rainforest—see Wurster et al., 2010; Xhaufclair et al., 2017) that would be open to human exploitation, only to be closed off when environmental circumstances varied (e.g. see Piper et al., 2011). Humans had to be adaptable to make the best opportunities of this constantly changing environment (Hunt et al., 2012; O'Connor et al., 2011; Samper Carro et al., 2016).

Petraglia et al. (2010: 301) discuss the impact of Late Pleistocene climate change on human populations living in the Indian Subcontinent, and with the exception of reference to desert environments their evaluation is equally applicable to SE Asia:

Arid phases in South Asia would have led to the expansion of the Thar Desert, but also the production of tropical forest refugia in southwest India and Sri Lanka, as well as the widespread creation of fragmented, mosaic grasslands and woodland habitats in most of the subcontinent. Small groups of foragers could have easily survived and pursued their hunting and gathering lifestyles in this range of habitats.

One aspect of this changing environment that had special significance in Mainland and island SE Asia was megafaunal extinctions, and Louys et al. (2007: 169) have shown how

... changes in sea level may have allowed easier routes for early colonising mammals, humans among them. It also brought about changes in the structure of the vegetation, disrupted river systems, and isolated islands like Java and Sumatra. This aspect of extinctions is unique to Southeast Asia—it is unlikely that changes in sea level affected the Americas, Europe or Australia in quite the same way.

That said, the role that humans played in these extinctions has still to be determined on a regional basis, or on an island-by-island basis in the Lesser Sunda chain of islands in eastern Indonesia.

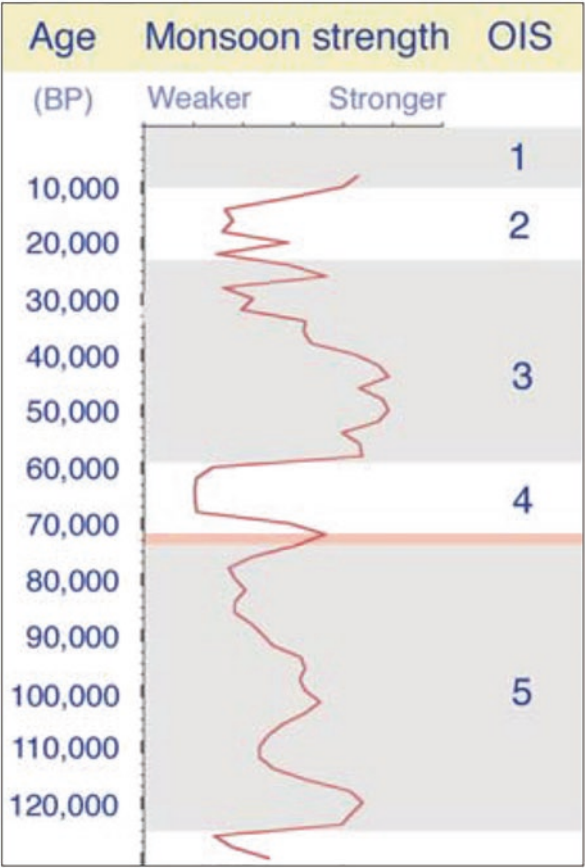


Fig. 24.40 Variations in rainfall over the past 130 kyr as reflected in the strength of the monsoon; 1–5 indicate Oxygen Isotope Stages (adapted from Petraglia et al., 2010: 298, Figure 2).

Some of these environmental and associated ecological changes would have impacted on the astronomical systems of the Australo-Melanesians, and our task as ethnoastronomers is to try and identify and quantify these.

24.9.2.2 Australo-Melanesians, the Post-Glacial Sea Level Rise and Evolving Astronomical Systems

The most dramatic changes in sea levels experienced by the Australo-Melanesians occurred during terminal Pleistocene times, between 22 kyr and 6 kyr when the seas continued their rapid and relentless rise towards current levels, drowning habitat after habitat as they flooded the Sunda shelf (Hanebuth et al., 2000; Sathiamurthy and Voris, 2006), destroying Sundaland in the process (Oppenheimer, 1998), and

forming the present-day islands of island Southeast Asia (e.g. for the case of Palawan, see Robles et al., 2015). Soares et al. (2008: 1215–1216) suggest that these sea level changes

... triggered major displacements of human groups living on the Sunda coastline and had an important role in shaping subsequent life in the region, in particular its maritime orientation and development of sailing technology ...

With the opening up of approximately twice as much coastline as there had previously been, the coastal populations of Sundaland (and perhaps also northwestern Wallacea) would have been well placed to expand into this rapidly growing habitat. Conversely, the hinterland-based populations would have faced severe land pressures from both the drowning of their ancestral lands and the encroachment of rainforest over the savannas and monsoon forests to which they had adapted (Bird et al. 2005).

There is accumulating archaeological evidence of the ways in which the occupants of Southeast Asia responded to these environmental changes (e.g. see Lewis et al., 2008b; Mijares, 2008; Pawlik et al., 2014a, 2014b), but not all environmental changes demanded major ecological changes. Sometimes biomes remained the same during the sea level rise—as in the case of mangrove forests (Berdin et al., 2003)—and it was merely a matter of redefining the boundaries of the changing human exploitation territory.

Although it is clear that populations were split and became geographical bound as a result of the post-glacial sea level changes, maritime technology allowed continuing contact between separated populations, thereby facilitating trade (e.g. see Reepmeyer et al., 2011), parallel cultural evolution and the perpetuation of evolving astronomical knowledge systems. Some of the Late Pleistocene genetic evidence assembled by Hill et al. (2007) and Soares et al. (2008) can be interpreted in terms of these sea level changes and the relocation of human populations.

As an aside we should mention that in Australia some coastal Aboriginal groups have a memory of this late-Pleistocene sea level rise preserved in their myths and legends (see Nunn and Reid, 2016), and it will be interesting to see if comparable accounts emerge from any island Southeast Asian negrito groups.

24.9.2.3 The Arrival and Superimposition of Austronesian Astronomical Systems

Without doubt, one of the most important threats to the long-term survival of the established astronomical systems of the Australo-Melanesians occurred about 4000 years ago when a totally new human population swept into island South East Asia. Termed the Austronesians, these people of Asian genetic stock had their origin in the farming communities of south-eastern China (Chi and Hung, 2010; Lu, 2006), but were firmly settled in Taiwan (Ko et al., 2014) before finally making their way southwards into northern Luzon, via the Batanes Islands (Bellwood and Dizon, 2008; Piper et al., 2009). This population movement is not only documented in the

archaeological record, but is clearly identified through genetic studies (e.g. see Kayser et al., 2008; Lipson et al., 2014; Mörseburg et al., 2016; Tabbada et al., 2010; Trejaut et al., 2005, 2014).

One of the features of the Austronesian settlement of island Southeast Asian that has preoccupied some scholars is the timing, given that the Austronesians were firmly (and apparently happily) settled in Taiwan for perhaps 1500 years before they began to move southwards and occupy Luzon and other islands in the Philippine archipelago (Bellwood, 2005). It transpires that the time when this migration occurred was marked by extremely low temperatures (see Berkelhammer et al., 2012), coupled with several hundred years of alternating drought and flood conditions as the monsoon was dominated by the El-Niño-Southern Oscillation (ENSO). This corresponds to what geologists and paleoclimatologists refer to as the '4.2 Ka Event', and it has been investigated in detail by Kathayat et al. (2018: 1875):

(Fig. 8) illustrates the ISM [Indian Summer Monsoon] variability between ~3.8 and 4.6 ka ... The interval marking the onset of the 4.2 ka event in our record (~4.255 ka) is indicated by a transition from pluvial (inferred by the lower $\delta^{18}\text{O}$ values) to variable ISM (dry-wet) conditions, with the latter superimposed by a few short-term (<decade) droughts (Fig. 8). Subsequently, the period between 4.07 and 4.01 ka is marked by persistently lower $\delta^{18}\text{O}$ values, implying stronger ISM (Fig. 8). The latter was terminated by a rapid increase in the $\delta^{18}\text{O}$ values (~1.0‰, Fig. 5), suggesting an abrupt weakening of the ISM at ~4.01 ka that occurred within a period of ~10 years. Notably, as discussed above, the ML.1 and ML.2 $\delta^{18}\text{O}$ profiles show gradual increasing trends over the entire length of the record, which was punctuated by two multi-decadal weak monsoon events centered at ~3.970 (~20 years) and ~3.915 ka (~25 years), respectively (Fig. 8). These aspects of our ISM reconstruction differ from previous proxy records from the ISM domain, which typically portray the 4.2 ka event as a multi-century drought (e.g., Berkelhammer et al., 2012; Dixit et al., 2014). Our new data, however, demonstrate that prominent decadal to multi-decadal variability, together with the intermittent occurrence of multi-decadal periods of low rainfall, was the dominant mode of ISM variability during the period coeval with the 4.2 ka event ...

The Fig. 8 referred to above by Kathayat et al. (2018) is reproduced here as Fig. 24.41. Perhaps it was the never-ending succession of floods and long-term droughts experienced in Taiwan from around 3,800 BP, coupled with increasing population pressure on land, which triggered the initial migration of Austronesians to the Philippines.

One of the mysterious features of this migration is that although the Asian mainland newcomers were experienced horticulturalists, there is little evidence that rice or millet-farming or their traditional cultigens (banana, breadfruit, coconut, pineapple, taro and yam), were important once they settled in Luzon (Bulbeck, 2008), and all of the early archaeological evidence indicates that they practiced broad-spectrum hunting and gathering where they made optimal uses of the various biomes within their exploitation territories (see Orchiston et al., 2021b). Only perhaps 1500 years later did rice-cultivation become important (Barker and Richards, 2013; Snow et al., 1986). The astronomical implications of these ecological changes warrant investigation.

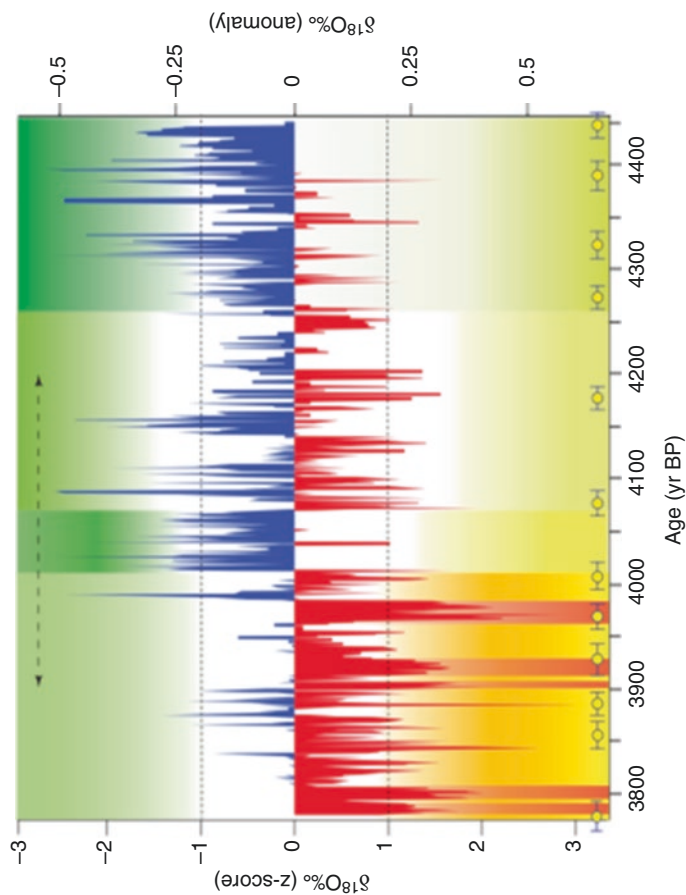


Fig. 24.41 The inferred pattern of Indian summer monsoon variability during the 4.2 ka Event. The horizontal dashed double arrows show the commonly-accepted duration of the 4.2 ka Event. Periods with yellow shading experienced drier conditions while those in green had wetter conditions. The red bars show periods of multi-decadal drought, and the blue bars intervals of intensive rainfall (after Kathayat et al., 2018: 1876).

The present-day distribution of negrito ethnic groups in the Philippines (Fig. 24.42) would indicate that the Austronesians must have frequently encountered negritos when they first occupied Luzon, and genetic data point to extensive initial gene-sharing between these two populations (e.g. see Scholes et al., 2011), followed in some cases by isolation (Heyer et al., 2013). However, gene-sharing was a long-term process among the Mamanwa of Mindanao, as illustrated in Fig. 24.43. Although we have no details of the dynamics of these interactions we note that initially the Austronesians mimicked the ecological practices of the local negrito groups, while the negritos eventually ended up adopting the language of the newcomers (see Reid, 1987, 1994).

The Austronesians did not just occupy the Philippines area but they also settled in the eastern half of Indonesia (Gomes et al., 2015; Mona et al., 2009; Spriggs, 2011), and then some groups exploited the advanced maritime technology that was a feature of their culture to venture eastwards (Tumonggor et al., 2013), establishing settlements on and off the coasts of New Guinea (see Skelly et al., 2014), then in Near Oceania (Friedlaender et al., 2007; Spriggs, 2000), before eventually spreading across the Pacific as the Polynesians (Bellwood, 2017; Kayser et al., 2008). Notwithstanding the passage of time, one interesting research project will be to determine whether any remnants of the ancestral Austronesian astronomical system of island Southeast Asia have survived in Polynesian astronomy. An equally insightful study will be to see which elements of Austronesian astronomy were adopted by the negrito populations of the Philippines, and *vice versa*.

24.9.2.4 The Arrival and Superimposition of Austro-Asiatic Astronomical Systems

Archaeological (Bellwood, 2017), skeletal, and genetic evidence (Hill et al., 2007; Karafet et al., 2010; Lipson et al., 2018; Matsumura et al., 2011, 2018, 2019; McColl et al., 2018) document the arrival of a second major influx of Asian settlers in island Southeast Asia, beginning about 3000 years ago. These immigrants, known as Austro-Asiatics, originated from South China and Vietnam, and are thought to have entered island Southeast Asia following a land route that ran from Vietnam to Java, via Thailand and the Malay Peninsula and Sumatra. From Sumatra they spread northwards into Borneo, and from Java they migrated northwards into Sulawesi (Simanjuntak, 2017) and eastwards into the Lesser Sunda Islands. However, "... in most areas of Southeast Asia where they settled (except in some parts of the Malay Peninsula) the Austro-Asiatic groups decided to abandon their original mainland Southeast Asian language and adopt the Austronesian languages that already were widely spoken and accepted throughout Island Southeast Asia." (Orchiston et al., 2021b: 65).

These latest newcomers to island Southeast Asia were the ancestors of the majority of the current occupants of island Southeast Asia. They were

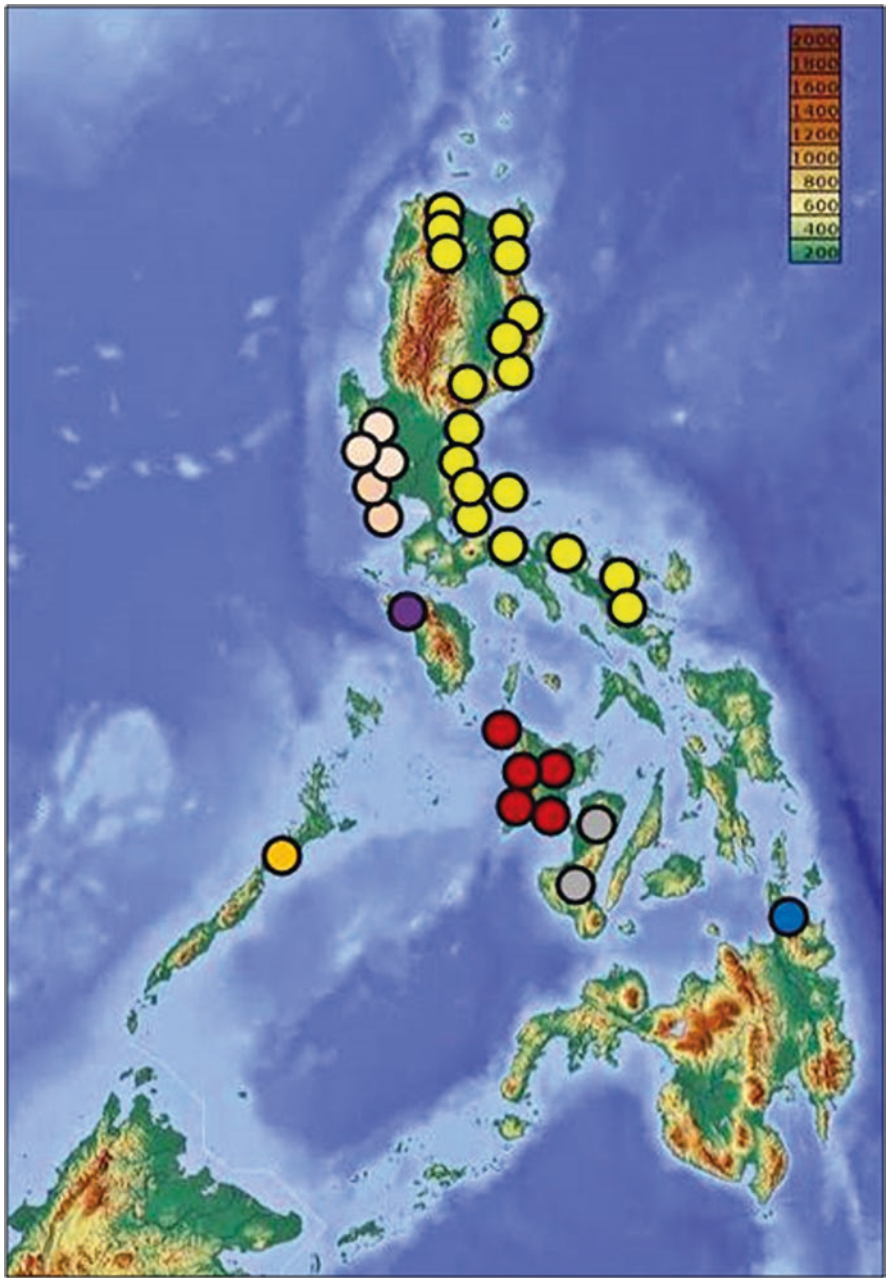


Fig. 24.42 The geographical distribution of the main Philippine negrito groups (Aeta = pink; Agta = yellow; Ata = grey; Ati = red; Batak = orange; Iraya = purple; Mamanwa = blue; map: Wayne Orchiston).

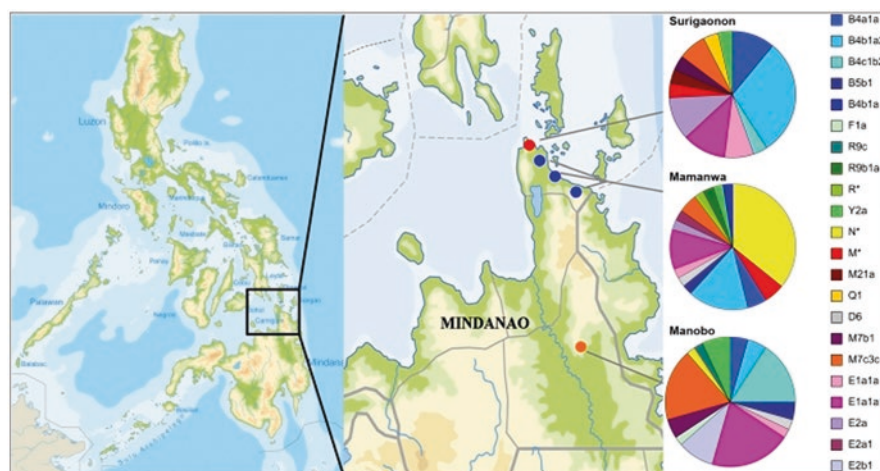


Fig. 24.43 A diagram showing the main haplogroups of the Mamanwa negritos of northern Mindanao and the extent of gene-sharing by the neighbouring Manobo Austronesian ethnic group (after Gunnarsdóttir et al., 2011: Figure 3).

horticulturalists, experienced in rice-cultivation, and *Pranotomongso* of Java (Dardjoeni and Hidayat, 1987; see Fig. 24.44) and a comparable rice-growing calendar found in northwestern Malaysia (Jaafar and Khairuddin, 2021b) reflect local modifications to the ancestral Austro-Asiatic astronomical systems, and evolved in island Southeast Asia in response to specific cultural and environmental demands (Orchiston and Orchiston, 2018). For example, we see the modification of these astronomical systems displayed in the origins of the months shown in Fig. 24.44: most of these are of Indonesian derivation, but two, Dhestal and Sadaha, are from India. For further details of the *Pranotomongso* calendar see Suryadarma (2019).

In addition to developing their own distinctive components of their astronomical knowledge bases, throughout island Southeast Asia the Austro-Asiatic groups had to decide which elements—if any—of the astronomical systems of the existing populations they should adopt. This was a particular challenge in what is now Indonesia where they encountered vibrant Austronesian *and* negrito (i.e. Australo-Melanesian) astronomical systems, as reflected in the ancestry bars in Fig. 24.45 that show combinations of green, blue and orange, or green, blue and yellow. On the other hand, our challenge as ethnoastronomers is not just to unravel these cultural changes, but also to determine the extent to which the Austronesian and negrito astronomical systems in this southern region of island Southeast Asia adopted, or were inspired by, elements of the newly-introduced astronomical systems of the Austro-Asiatics.

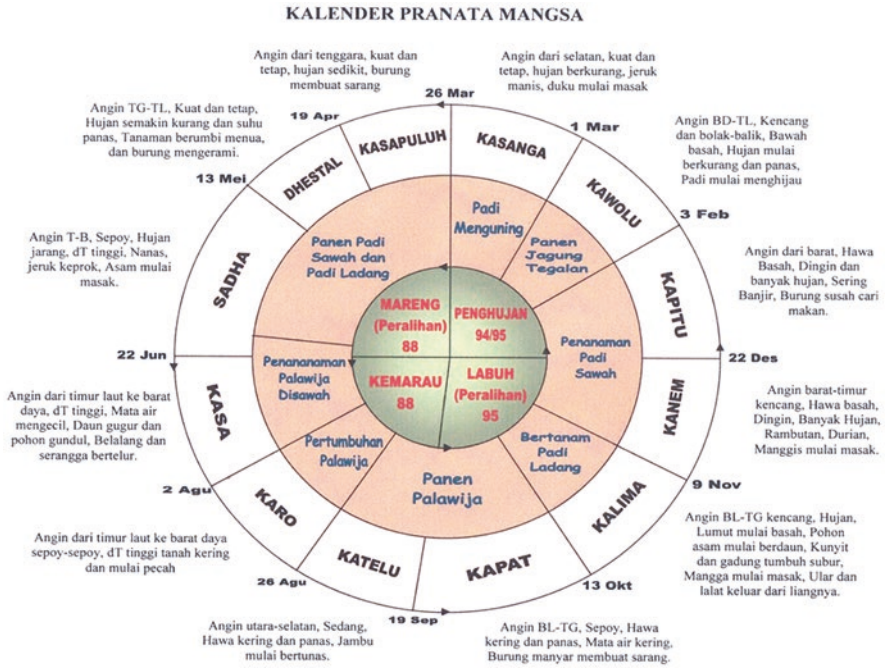


Fig. 24.44 The *Pranotomongso* (rice-growing) calendar of Java, Indonesia. The four central green sectors show each of the ‘seasons’ (anti-clockwise from the top: transition, dry, transition, and wet, and the duration, in days of each season). The eight pink sectors outline the sequence of planting, cultivating and harvesting the rice (anti-clockwise from the top: harvest lowland and upland rice; plant new seedlings in the rice fields; the plants grow; harvest the rice; plant new seedlings in the rice fields; tend the crops; harvest the corn; yellow rice). The twelve white sectors show the months of the year in the *Pranotomongso* calendar. The outer information describes the wind, temperature, rainfall and other characteristics of these periods of the year relative to rice cultivation, as summarized in Table 24.1 (after Kearifan lokal kalender tanam ..., 2018).

24.9.2.5 The Spread of the Hindu and Islam Religions throughout Island SE Asia and Changing Astronomical Systems

Hindu religion spread throughout much of mainland and island Southeast Asia from about 1200 BP (see Fig. 24.46), and had a profound impact on Southeast Asian astronomical systems (as outlined in Section 24.6 above). It was associated with both the transmission of ideas *and* people, but the ‘people’ in this case only involved small groups—religious leaders and their supporters—not large-scale mass migrations spread over hundreds of years, as is thought to have occurred when the Austronesian and Austro-Asiatic populations settled island Southeast Asia.

The second major religion to spread throughout Southeast Asia was Islam, beginning about 800 years ago (see Fig. 24.47), but it was mainly associated with the

Table 24.1 Characteristics of the months of the year in the *Pranotomongso* calendar (after Kearifan lokal kalender tanam ..., 2018).

Time Period	Month	Characteristics
22 June–1 August	First	Winds are from the northeast to the southwest. Temperatures are high. Springs dry up. Leaves fall and trees are bare. Grasshoppers and insects are seen.
2 August–25 August	Second	Winds are from the northeast to the southwest and are strong and continuous. Temperatures are high. The soil is dry and starting to break up.
26 August–18 September	Third	Winds are from the north to the south and are moderate. It is dry and hot. Guavas begin to ripen.
19 September–12 October	Fourth	Winds are from the northwest to the southeast and are continuous. It is dry and hot, and springs are dry. Weaver birds make their nests.
13 October–8 November	Fifth	Winds are always from the northwest to the southeast. It is raining and moss is growing. Turmeric and gadung thrive, and mangoes start to ripen. Snakes and flies appear.
9 November–21 December	Sixth	Winds are from the east to the west and are strong. It is wet and cold, with heavy rain. Rambutan and mangosteen start to ripen.
22 December–2 February	Seventh	Winds are from the west. It is wet and cold, with heavy rain. There is often flooding. Birds have difficulty finding food.
3 February–28/29 February	Eighth	Winds are always from the southwest to the northeast and are variable. It is wet and hot, but the rain starts to lessen. The rice starts to turn green.
1 March–25 March	Ninth	Winds are from the south, and are strong and regular. There is less rain. Oranges begin to ripen.
26 March–18 April	Tenth	Winds are from the southeast and are strong and steady. There is little rain. Birds make their nests.
19 April–12 May	Eleventh	Winds are from the southeast to the northeast and are strong and steady. There is even less rain and temperatures become hotter. Crops ripen. Birds lay their eggs.
13 May–21 June	Twelfth	Winds are from the east to the west and are continuous. Rain is rare, and temperatures are high. Pineapples, tangerines and tamarind begin to ripen.

transmission of ideas and belief systems, not wholesale migrations of people. While Hindu influences permeated many aspects of everyday life, Islam was reserved mainly for religious observances (see Section 24.7, above). In this context, sometimes in rural areas the traditional astronomical systems introduced originally by the Austro-Asiatcs were allowed to continue without alteration, but at other times they were modified in order to accommodate Islamic concepts, or were abandoned entirely. Thus, when she did ethnoastronomical fieldwork on the island of Madura (off the north-east coast of Java) in 2015–2016, Siti Fatima found, particularly in the towns, that “... some people now prefer to respond to eclipses in a traditional Islamic way through *shalat* (prayers), even if they are aware of the traditional Madurese responses expected during eclipses.” (Fatima et al., 2021: 583).

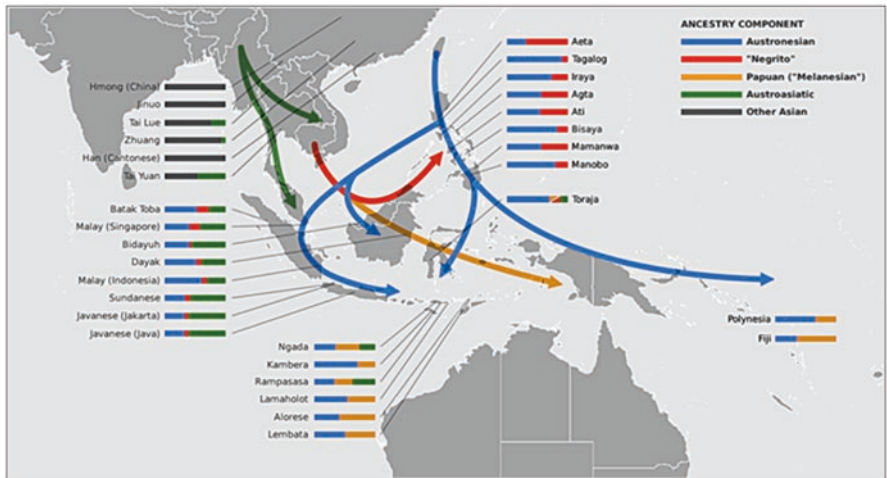


Fig. 24.45 Genetic evidence from island Southeast Asia that clearly identifies the presence of the Australo-Melanesian (orange and red), Austronesian (blue) and Austro-Asiatic (green) ancestral populations. Their genetic imprints are now preserved in the present-day populations of island Southeast Asia (after Lipson et al., 2014: Fig. 2).



Fig. 24.46 A map showing the spread of the Hindu religion (sometimes in tandem with Buddhism) from the Indian Subcontinent throughout mainland and island Southeast Asia in the last 1,200 years (https://commons.wikimedia.org/wiki/File:Hinduism_Expansion_in_Asia.svg).



Fig. 24.47 A map showing the spread of Islam into island Southeast Asia during the thirteenth to sixteenth centuries (xenohistorian.wordpress.com).

Table 24.2 Colonialism in Southeast Asia (adapted from Ty, [n.d.](#)).

Colonising Nation	Territory Occupied	Time Interval
Portugal	Throughout island SE Asia initially, until expelled by the Dutch in the seventeenth century, leaving only Timor as a Portuguese colony.	1511–1641/1975
Spain	The Philippines archipelago.	1565–1898
Netherlands	The Dutch East Indies (most of present-day Indonesia).	1605–1799; 1825–1940s
Great Britain	Burma (now Myanmar) until independence in 1948; Singapore; Malaya (now Malaysia) until independence in 1957.	1824–1957
France	Cochin China (now Laos, Cambodia and Vietnam) until independence in 1954.	1859–1954
United States of America	The Philippines archipelago, until independence in 1946.	1898–1946
Japan	Much of mainland and island Southeast Asia	WWII

24.9.2.6 The Spread of Western Colonialism throughout SE Asia

The Spanish had colonial ambitions in the Philippines archipelago and the Dutch in what is now Indonesia from the sixteenth century, in competition mainly with the British and the Portuguese, while the French tended to focus on mainland Southeast Asia (see Fig. 24.48). The United States of America came later (see Table 24.2). All of these colonising nations were inspired by commercial opportunities, international status and prestige, and also in the case of the Philippines the desire by Catholic priests to win religious converts (see Hall, 1981).

Apart from their involvement in eclipse expeditions to mainland and island Southeast Asia, as summarized above in Section 24.8, astronomical interests of the colonial nations were restricted mainly to the establishment of local time services,



Fig. 24.48 This colourful map shows the territories of the different colonial powers in Southeast Asia. Code: red, purple and blue were British (Burma = Myanmar; Malaya, Sarawak and British North Borneo = Malaysia, and Singapore); green (top-left) was French (Indochina = Cambodia, Laos and Vietnam); orange was Dutch (Dutch East Indies = Indonesia); yellow was Spanish up to 1898 and American thereafter (the Philippines—the only colony shown here that subsequently did not change its name); and dark green (bottom, right of centre) was Portuguese (Timor) (<https://www.ehm.my/about/history-of-malaysia>).

and the gathering and dissemination of meteorological data. Accurate knowledge of time was critical for commerce and communications, and in the nineteenth and early twentieth centuries was best portrayed by the presence of time balls and other devices used to indicate time at various ports and other locations throughout Southeast Asia. The geographical distribution of these is shown in Fig. 24.49, and is discussed in the chapter by Kinns (2021) elsewhere in this book.

The focus on meteorology also was understandable given the agricultural and horticultural concerns of the colonial nations, and the very real threats posed by droughts and cyclones (for the Philippines see García Herrera et al., 2007). Thus, for the most part the astronomical observatories that were set up were small-scale affairs (Williamson 2020)—sometimes nothing more than a transit telescope facility with an astronomical clock, connected to a time ball. That said, looks can be deceptive because even the very impressive-looking Central Indochina Observatory in Haiphong (northern Vietnam) shown in Fig. 24.50 (Lagrula, 1935; Pham and Le, 2021) only housed an extremely modest 12-cm (4.7-inch) refracting telescope. Meanwhile, Manila Observatory in the Philippines (Fig. 24.51) could boast a 48.3-cm (19-inch) refractor (Seitzer, 2020)—by far the largest in Southeast Asia at the

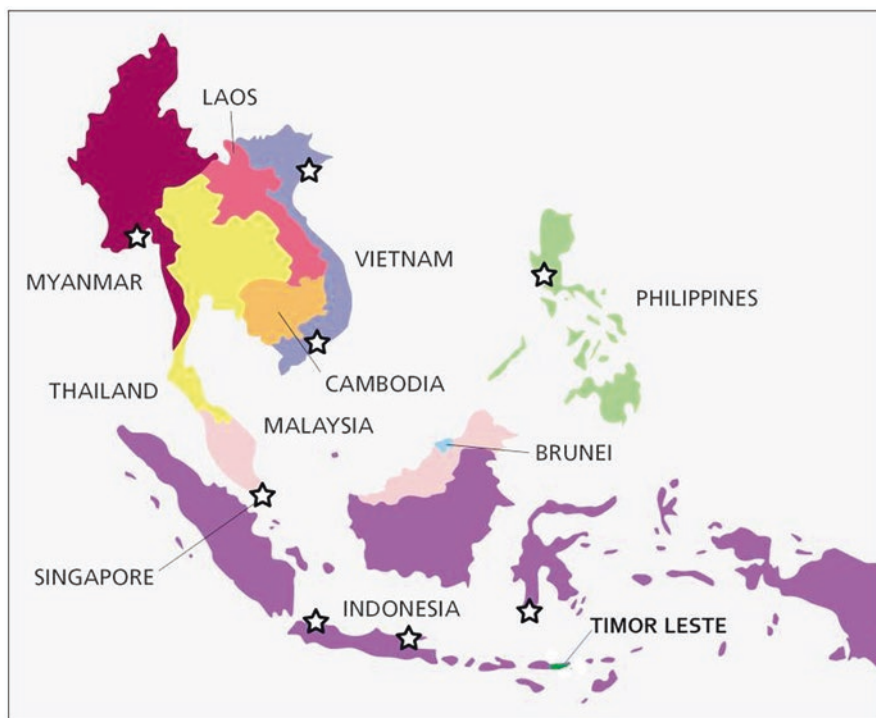


Fig. 24.49 The white stars show the locations of the time balls of mainland and island Southeast Asia (after Kinns, 2021). From north to south, they were at Haiphong, Rangoon, Manila, Hanoi, Singapore, Makassar, Jakarta and Surabaya (map: Wayne Orchiston).



Fig. 24.50 The Central Indochina Observatory, in Haiphong, which was established by the French in 1902 (photograph courtesy: Klaus Hülse Collection).



Fig. 24.51 A view of Manila Observatory. The 19-inch refractor with a lens by Merz (Germany) and mounting by Saegmuller (USA) was housed in the dome on the extreme left (courtesy: Manila Observatory Archives).

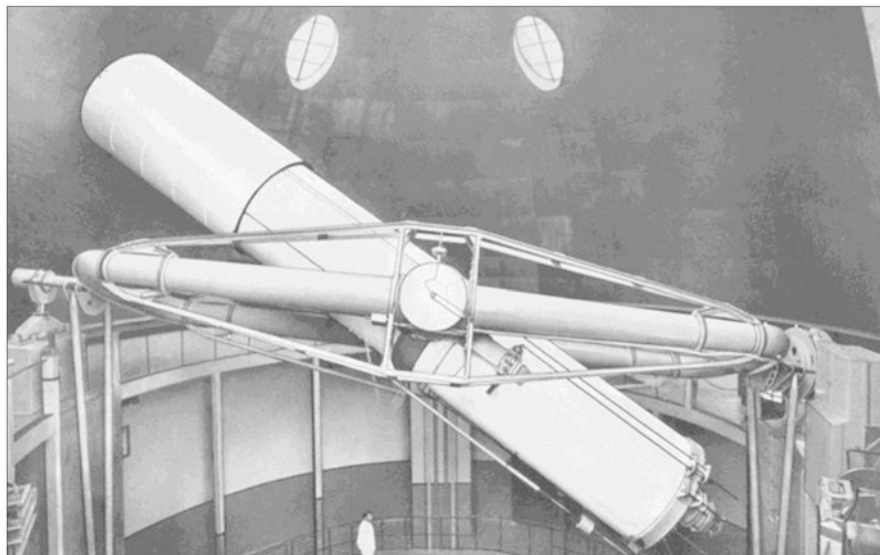


Fig. 24.52 The 60-cm Zeiss double refractor (after Voûte, 1933: Plate V).

time—but Manila’s tropical weather meant that its research potential was limited. So the professional astronomical observatories of Southeast Asia were totally unlike those maintained during the nineteenth and early twentieth centuries in other Asian and Oceanic British colonies such as India and Australia, which also were actively engaged in astronomical (and later astrophysical) research (e.g. see Haynes et al., 1996; Kochhar and Orchiston, 2017; Orchiston, 2017: Chapter 3).

Another notable feature of Southeast Asia’s professional astronomers was that they had little if any desire to impart astronomical knowledge to the indigenous populations, or address the perceived short-comings of local astronomical knowledge systems—unlike in India where Indian Assistants such as Madras Observatory’s Ragoonatha Charry were very active in trying to raise the level of astronomical literacy among local ethnic groups (see Shylaja, 2012; Venkateswaran, 2019).

Nor did a vibrant amateur astronomy tradition—as observed in the British colonies of India (Kochhar and Orchiston, 2017), Australia (Orchiston, 2017 Chapter 4) and New Zealand (Orchiston, 2016)—ever emerge in mainland or island Southeast Asia during the nineteenth and early twentieth centuries, although in the Dutch East Indies there was a collaboration between the professional astronomer, J.G.E.G. Voûte (1879–1963) and the wealthy Dutch tea plantation owner and astronomy enthusiast, K.A.R. Bosscha (1865–1928) that led to the founding of Bosscha Observatory in 1923 (see Hidayat et al., 2017: 335–338; Voûte, 1933: 38). This was located at Lembang, in western Java, and the primary instrument was a 60-cm (24-inch) twin refractor by Zeiss (see Fig. 24.52).

Thus far our focus has been on professional European- and American-style astronomy and the associated astronomers, but we should not forget that many Jesuit missionaries were trained in mathematics and astronomy, and were experienced astronomical observers. As we noted in Section 24.8 Jesuit missionary/

astronomers introduced Western astronomy to Siam (present-day Thailand) during the seventeenth century, but their stay was short-lived (Orchiston et al., 2021a, 2021d). Early Jesuit astronomical observations also were made from what is now Vietnam (see Bonifácio and Malaquais, 2018), and evidence of such observations from the Philippines and Indonesia is now the subject of on-going studies, aided in the case of the Philippines by Blair and Robertson's (1903–1909) invaluable 55-volume work, *The Philippine Islands, 1493–1898*. This is based on archival records in Europe, and documents the history of the Jesuits in the archipelago (for the astronomical content see Dela Crux, 2020). Finally, nor should we forget that the aforementioned Manila Observatory was set up by the Jesuits in 1865 as part of their world-wide network of observatories (see Udias, 2003), only to be destroyed during World War II.

24.9.3 *Two Conspicuous Southeast Examples of Multiplex Astronomy (YÔ and WO)*

'Multiplex astronomy' was common throughout mainland and island Southeast Asia, where successive ethnic groups followed by religions superimposed their own astronomical imprints on the original astronomical systems first introduced by the Australo-Melanesians. In completing Section 24.9 we will present just two case studies that perfectly demonstrate 'multiplex astronomy'. Both are from mainland Southeast Asia.

24.9.3.1 *The Cham Calendar of Vietnam*

There was a Champa Kingdom in central and south Vietnam from the late second century CE through to 1832 (see Fig. 24.53), where they had an Indianized calendar that used the Śaka Era. There also are inscriptions using the Śaka Era that date from the sixth century, so we know that the Indianized calendar was used from quite an early date in Southeast Asia (see Sugimoto, 1956).

The Cham people in present-day Vietnam are said to be the descendants of the Champa Kingdom, and they have a rather interesting calendrical system (Nakamura, 1999, 2009; Yoshimoto, 2000, 2003, 2011) that has been intensively studied in recent years by Professor Trương Văn Món (see Sakaya, 2016). The Cham 'calendar' actually comprises two different calendars:

- (1) The '*Ahier*' Calendar: This is an Indianized luni-solar calendar, but the 12-year Chinese cycle with animal names is also used.
- (2) The '*Awal*' Calendar: This is an Islamic lunar calendar, which was introduced in the tenth century.

These two types of calendar are used by the Cham people in Vietnam. This is an interesting example of multiplex astronomy.

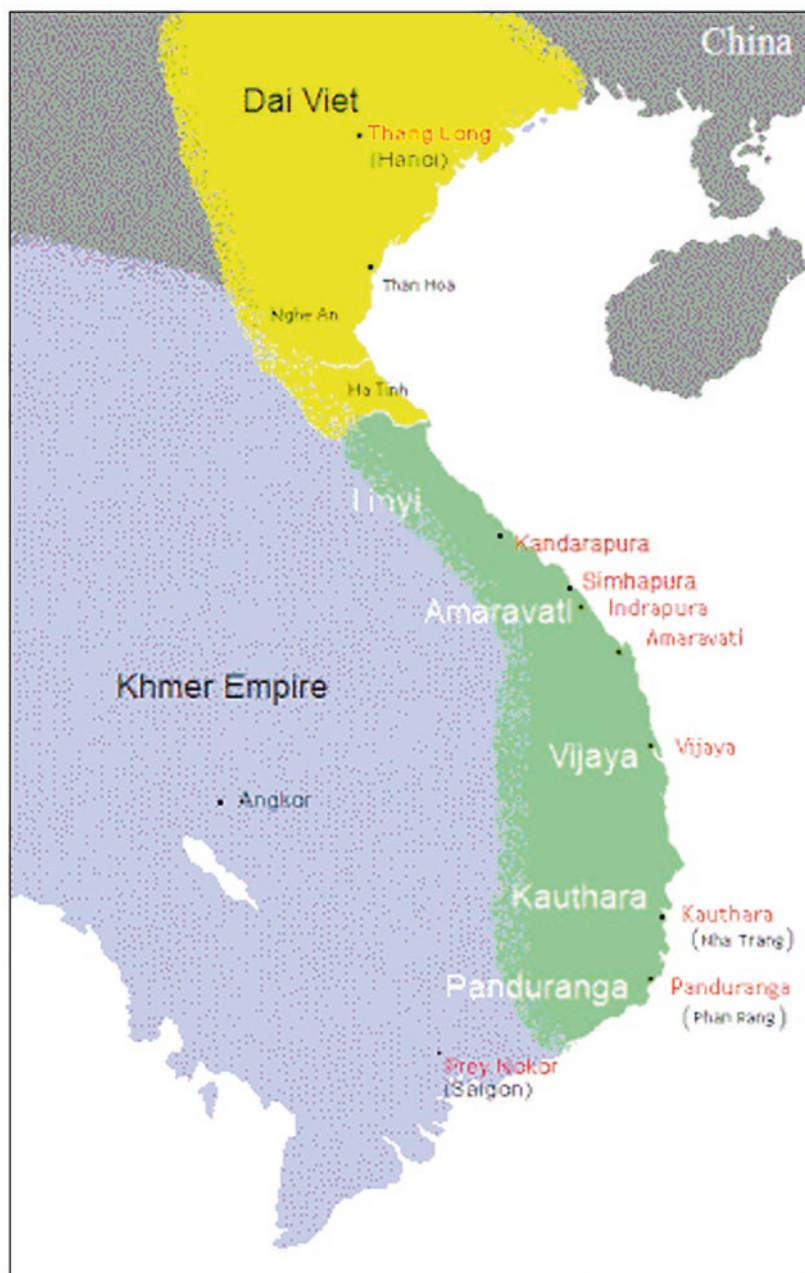


Fig. 24.53 A map showing the extent of the Champa Kingdom (green) between CE 1000 and 1100, and the main population or religious centres (in red) (<https://en.wikipedia.org/wiki/Champa#/media/File:VietnamChampa1.gif>).

24.9.3.2 The Traditional Thai Calendar

One year in the traditional Thai calendar is a sidereal year, that is, 365.25875 days. This is similar to a year of the Indian Ārdharātrika school, and is evidently of Indian origin. However, the Chinese 12-year cycle of animals is also used in the traditional Thai calendar. And, also, the use of serial numbers to denote lunar months may be of Chinese origin.

For intercalation, a kind of simplified method using the 19-year and possibly the 57-year ($57 = 19 \times 3$) cycle is used. For the 19 years, 7 intercalary months are added. The 19-year cycle of intercalation is mentioned in the ‘Souriat’ of the Ayutthaya Dynasty (de la Loubère, 1693(II): 190) and was quite popular in ancient China, but was not popular in ancient India.

If the 57-year cycle is strictly used, 57 years are 705 months ($705 = (19 \times 12 + 7) \times 3$) and are 20,819 days ($20819 = 6 \times 57 \times (29 + 30) + 7 \times 3 \times 30 + 11$). Then, one year becomes 365.2456 days, and one lunar month becomes 29.530496 days. Here, one year is close to a tropical year.

This 57-year cycle is not found in Indian calendars or in official Chinese calendars. However, there was a modified *Taichu* (太初) calendar, which was not officially used, in the Eastern Han Dynasty (CE 25–220) of China, which has the same Southeast Asian 57-year intercalation. So, the first author of this chapter has hypothesised (Ôhashi, 2002, 2006a, 2011) that the Southeast Asian intercalation originated from the Chinese modified *Taichu* calendar (see Fig. 24.54 for the relationship between the length of a year and the length of a month. The line is horizontal when the 19-year cycle (or its multiple) of intercalation is used.

Eade (1995: 56) mentions that the 57-year cycle consists of 20,819 days, but it seems that the actual use of this cycle in Southeast Asia is uncertain. The cycle that is usually used in Southeast Asia is the 703-month cycle, which consists of 20,760 days (instead of 20,819 days of the 705-month cycle). Zhang and Chen (1981: 250–251) suggested the actual use of the 57-year cycle in the ‘Xitan’ calendar system of the Dai people in Yunnan Province, China. This topic needs to be researched further.

A similar calendar to the Thai calendar is in use in Myanmar, Laos and Cambodia, but in the actual calendars, some modifications have been made in different areas (including in Thailand) and in different ways. Gislén (2018) explains the difference between the Thai and Burmese calendars, and he suggests the possibility (Gislén, 2019) that the original Burmese calendar was influenced by the *Romaka-siddhānta* (one of the five astronomical systems described in the *Pañca-siddhāntikā*, an Indian Sanskrit text by Varāhamihira from the sixth century), and only later was the Hindu sidereal year adopted. This suggestion needs further investigation.

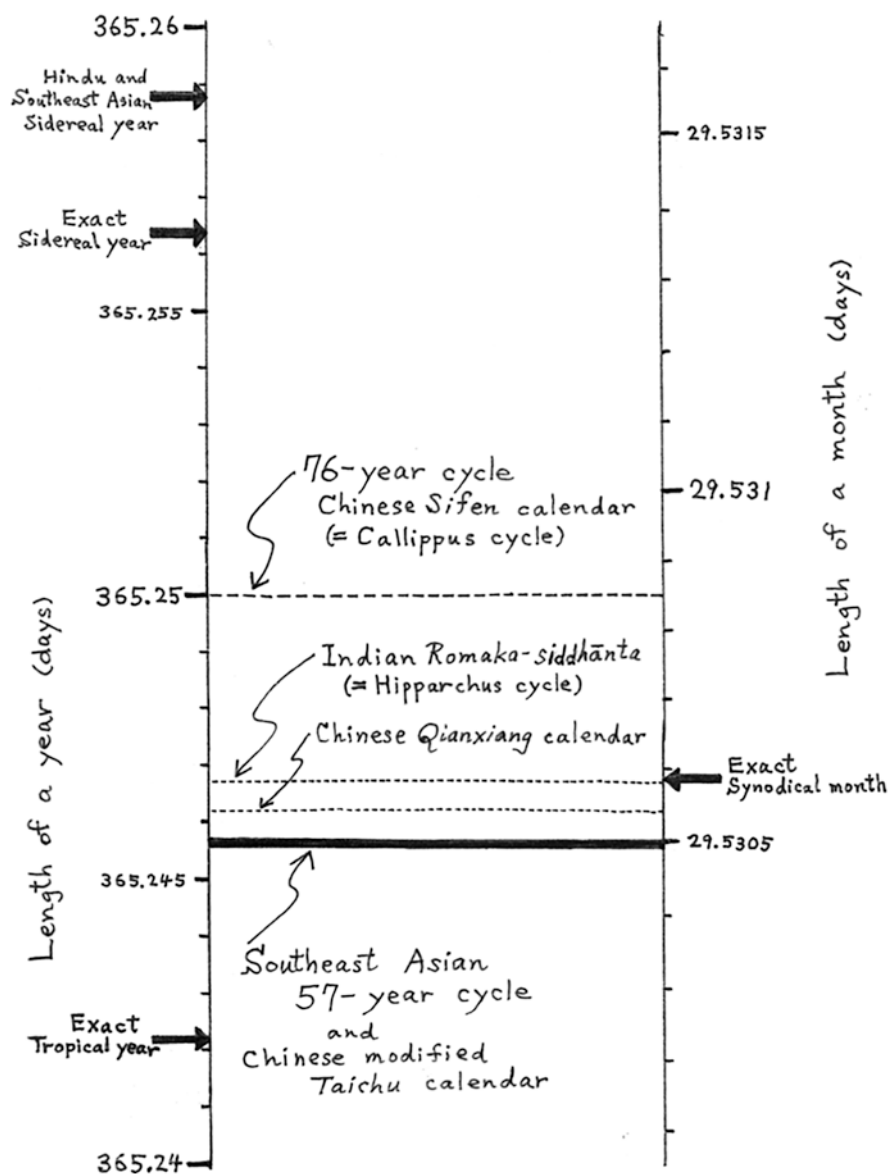


Fig. 24.54 The length of a year and a lunar-month when the 19-year cycle (or multiples of it) is used (diagram: Yukio Ôhashi).

24.10 Concluding Remarks (YÔ and WO)

Several different astronomical traditions are found in Southeast Asia that are regionally unique. For example, traditional Thai astronomy and some other Mainland Southeast Asian astronomical systems were created by combining local astronomical knowledge, Chinese influence and Indian influence, and they offer exciting research opportunities. Meanwhile, an even greater challenge lies in trying to identify the components of the various existing astronomical systems found throughout island Southeast Asia, given the complicated occupation history of this region and the key role that environmental change probably played. In this light, Southeast Asian astronomical history represents a unique field laboratory with a seemingly endless succession of challenging research projects for astronomical historians, ethnoastronomers, and especially graduate students.

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The first author of this chapter (YÔ) wishes to thank delegates from Indonesia, Malaysia, Thailand, the Philippines and Vietnam who attended the Ao Nang (2015) and Mandalay (2017) meetings of the SEAN History & Heritage Working Group and provided invaluable information that he was able to include in this chapter. For example, at the Ao Nang meeting he discussed the Thai calendar with Mr Vorapon Maison, and was grateful to Mr Pisit Nitiyanant for kindly providing him with a copy of Phengkaew's (2009) book. Again, just on calendars, at the Mandalay meeting YÔ learnt much about Vietnamese astronomy from the presentation by Associate-Professor Lê Thành Lân and Mr Phạm Vũ Lộc, and Mr Phạm also supplied details of the research on the Cham calendar by Professor Trương Văn Món (see Sakaya, 2016) and kindly provided him with a copy of his book. In addition, in the course of reading books in Thai, YÔ was grateful to his Thai teacher, Ms Nataya Yamakanon, for her assistance.

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